

4

Domestic Electricity

In this chapter, we will learn about

- power and energy consumption of domestic appliances
- household wiring and its safety features
- alternating current

4.1 Power and energy consumption of domestic appliances

DSE goal

Determine power rating and operating current for appliances.

DSE exam

13(1A)Q33, 14(1B)Q8(a)–(c), 15(1A)Q29, 18(1B)Q8(a)

1 Power and mains voltage



I brought my foldable kettle to Tokyo last week. Why did it take much longer time to boil the water there?

This was because the kettle operated at a lower voltage in Tokyo.



Mains electricity is the electricity that you get from the wall socket.

The kettle is powered by the **mains supply**. The mains voltage is 220 V in Hong Kong. It may be set at different values at different places. Therefore, an appliance may consume different power in different places. The actual operating power and current depends on its resistance.

Example 1 An appliance operating at different voltages (I)

Cross-link

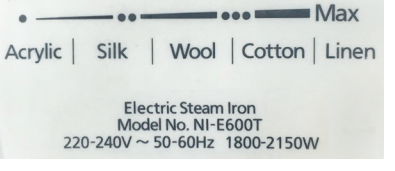
The rating '220 V, 1800 W' means that the power consumed by the iron is 1800 W when it operates at 220 V.

You learned about rated values in Ch 2 (p.96).

An electric iron is rated at '220 V, 1800 W'.

(a) Find the resistance of the iron.

(b) The iron is used in the USA where the mains voltage is the 120 V. Find the power consumed by the iron.



Solution

Recall these equations you learned on p.95.

$P = VI$

$P = I^2R$

$P = \frac{V^2}{R}$

(a) $R = \frac{V^2}{P} = \frac{220^2}{1800} = 26.9\ \Omega$

(b) $P = \frac{V^2}{R} = \frac{120^2}{26.9} = 535\ \text{W}$

mains supply 市電電源

In Hong Kong, the power ratings of appliances are usually based on an operating voltage of 220 V (Table 4.1a). If these appliances operate at other voltages, the actual power consumed will be different.

Source: HK Electric



Appliance	Typical rated power
Space heater	2000 W
Oven	1200 W
Air conditioner	1000 W
Refrigerator	120 W
TV	100 W

Table 4.1a *Typical power ratings of some appliances operating at 220 V.*

Google Forms Microsoft Forms

Quick check 1

→ E-version in OUP Exercise Platform is also available.

1 (a) The table below shows the rated values of appliances *X* and *Y*. Complete the table.

	V_0 / V	P_0 / W	R_0 / Ω
X	220	50	
Y		144	100

(b) The appliances in (a) now operate at voltages different from the rated value. Complete the table.

	V / V	P / W	R_0 / Ω
X	110		
Y		100	

2 An oven has a rated power close to

A 10 W.

B 100 W.

C 1000 W.

D 10 000 W.

3 *True or false:*

(a) For two appliances of the same rated voltage, the one with a higher rated power has a lower resistance. (T/F)

(b) When an appliance operates at a voltage lower than its rated value, the current through it is lower than the rated value. (T/F)

Learn by practice 1

1 The rating of a rice cooker is as shown. Find the power consumed if it operates at 110 V.



2 A heater is designed to work at 220 V. When it is connected to the 120-V mains, it consumes electrical power at a rate of 298 W. Find its rated power.

3 The power consumption of a kettle is 825 W when a current of 3.75 A flows through it. Its power consumption becomes 206 W when it operates at voltage *V*.

(a) Find *V*.

(b) Find the current flowing through the kettle.

4 When a lamp operates at 220 V, a current of 0.25 A flows through it. Find the current drawn and the power consumed if the lamp operates at 100 V.

Example 2 An appliance operating at different voltages (II)

The mains voltage in city A is 220 V and that in city B is 110 V. Compare the time taken to boil the same amount of water at the same initial temperature using the same electric kettle in the two cities.

Solution

$$E = Pt = \left(\frac{V^2}{R}\right)t$$

The same energy E is needed to boil the water. ▶

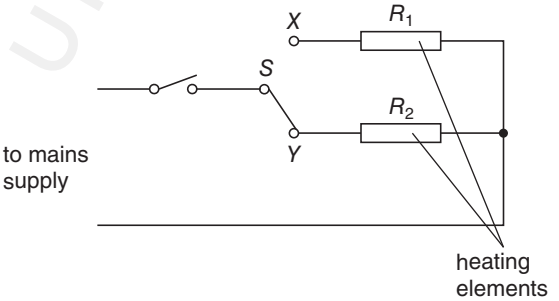
$$\therefore\ t = \frac{ER}{V^2} \propto \frac{1}{V^2} \text{ (for constant } E \text{ and } R)$$

$$\frac{t_A}{t_B} = \frac{V_B^2}{V_A^2} = \frac{110^2}{220^2} = \frac{1}{4}$$

$\therefore$ The time taken in city B is 4 times that in city A.

Example 3 An appliance operating at different voltages (III)

A heater is designed to operate at 220 V or 110 V with the same power of 1500 W. Its circuit is as shown. The resistances of the two heating elements are R_1 and R_2. Switch S should be connected to terminal X when the heater operates at 220 V and to terminal Y at 110 V. Find R_1 and R_2.



Solution

$$R_1 = \frac{V^2}{P} = \frac{220^2}{1500} = 32.3\ \Omega$$

Tip

To have the same P, by $R = \frac{V^2}{P}$, R should decrease when V decreases. ▶

$$R_2 = \frac{V^2}{P} = \frac{110^2}{1500} = 8.07\ \Omega$$

Google Forms Microsoft Forms

Quick check 2

→ E-version in OUP Exercise Platform is also available.

(For Q1–2.) An electric kettle is designed to operate at multiple voltages with the same power consumption.

1 Does the higher-voltage configuration have a higher or lower resistance?

2 The kettle takes time T to boil a pot of water when operating normally at the lower voltage. If its setting is changed to the higher-voltage configuration but is still operating at the lower voltage, how does the time taken to boil a pot of water change?

Learn by practice 2

1 An electric kettle rated '220 V, 1500 W' takes 6 minutes to boil a certain amount of water when connected to a 110-V mains supply.

(a) Find its resistance.

(b) Find its power consumption when working at 110 V.

(c) How long does it take to boil the same amount of water when working at its rated condition?

DSE goal

Use kW h as a unit of electrical energy.

Calculate cost of running appliances.

DSE exam

16(1A)Q28, 19(1B)Q8(c), 20(1A)Q2, 21(1B)Q7(b)

Caution

kW h is a unit of energy, not power.

2 Energy consumption

You pay the electricity company for using mains electricity. The electric bill is calculated according to the amount of electrical energy consumed, which is measured in **kilowatt-hours** (**kW h**) (Fig 4.1a).

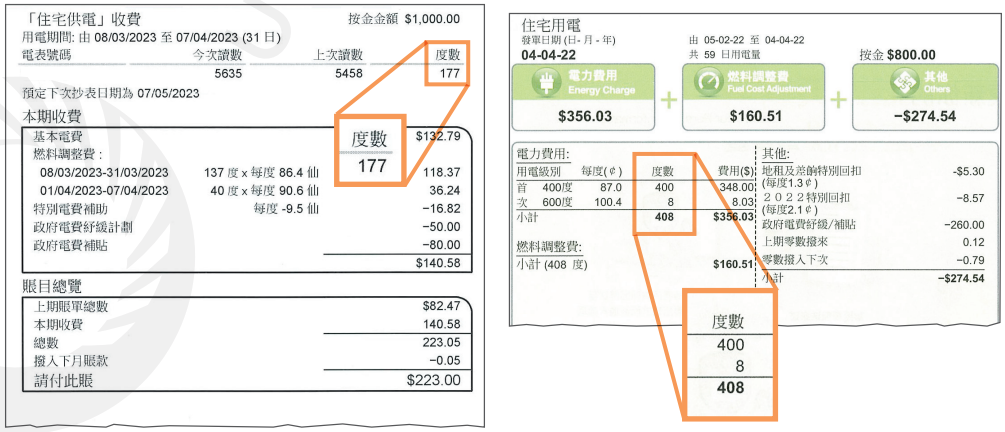


Fig 4.1a *Electric bills. The unit of electricity consumption is kW h.*

There is also fuel cost adjustment in the bill. Encourage Ss to read the bill and learn more from it.

In Hong Kong, the charge for every kW h of electrical energy increases with the amount of consumption. This is intended to encourage people to save energy.

What is the average cost of 1 kW h of electrical energy in each bill?

This unit is much larger than the joules (J).

$$1 \text{ kW h} = (1000 \text{ W}) \times (3600 \text{ s}) = 3.6 \times 10^6 \text{ J} = 3.6 \text{ MJ}$$

kilowatt-hour (kW h) 千瓦小時

Example 4 Energy consumed by heater

An electric heater consumes a power of 2 kW and is switched on for 45 minutes.



(a) Find the electrical energy consumed in (i) J and (ii) kW h.

(b) How much should be paid if 1 kW h of electricity cost \$1.1?

Solution

(a) (i) $E = Pt = (2000\ \text{W}) \times (45 \times 60\ \text{s}) = 5.4 \times 10^6\ \text{J}$

(ii) $E = Pt = (2\ \text{kW}) \times \left(\frac{45}{60}\ \text{h}\right) = 1.5\ \text{kW h}$

(b) Cost = \$1.1 × 1.5 = \$1.65

Tip
The energy should be in kW h when calculating the cost of electricity.

Google Forms Microsoft Forms

Quick check 3

→ E-version in OUP Exercise Platform is also available.

1 Which unit is different from the others?

A J B kW h

C W D W s

2 Express 50 000 J in terms of kW h.

A 0.0139 kW h

B 13.9 kW h

C 1.8×10^{11} kW h

3 The cost of electricity is \$1.1 per kW h. Complete the table.

	power	operating time	operating cost
(a)	0.5 kW	6 h	
(b)		30 min	\$0.9
(c)	200 W		\$18

Learn by practice 3

Take the cost of electricity as \$1.1 per kW h.

1 The power consumed by a TV is 100 W. Find the cost of electricity for operating the TV for 500 hours.

2 A 1500-W heater is operated for time T. The cost of electricity is \$135. Find T.

3 An air conditioner is operated 8 hours a day. The cost for operating it for 30 days is \$324. Find the power consumed by the air conditioner.

4 A 500-Ω lamp is connected to a 220-V mains supply. Find the cost of using the lamp for 100 hours.

Exercise 4.1

The cost of electricity is $1.1 per kW h.

*1 A cooker (Fig a) rated '220 V, 600 W' is operating at its rated condition.



Fig a

Which of the following statements is/are correct?

(1) Its resistance is 80.7 Ω.

(2) It draws a current of 2.73 A.

(3) It consumes 600 J of electrical energy per hour.

A (1) only

B (3) only

C (1) and (2) only

D (2) and (3) only

*2 A lamp is rated at '220 V, 100 W'. What is the cost of electricity for operating it at 110 V for 200 hours?

A $5.5

B $7.7

C $11

D $22

*3 Light bulbs L_1 of rating '220 V, 100 W' and L_2 of rating '220 V, 60 W' are connected in parallel to the 100-V mains. Find the power consumption of L_1 and L_2.

*4 Two light bulbs X and Y rated at '220 V, 60 W' and '220 V, 100 W' respectively are connected in series to a 220-V mains supply.

(a) Find the resistance of each bulb.

(b) Find the total power consumed by the bulbs.

(c) Are the bulbs brighter or dimmer than when they are operated at their rated values?

*5 The resistance of a lamp (Fig b) is 484 Ω.



Fig b

(a) Find the cost of operating it at 220 V for 1 hour.

(b) Compare the performance of the lamp when it works at 220 V to that at 110 V.

*6 When a heater is connected to a 220-V mains supply, a current of 6.8 A flows through it.

(a) Find the power consumed by the heater.

(b) The heater is then connected to the 110-V mains in another city.

(i) Find the current flowing through the heater.

(ii) Compare the power consumed by the heater in this city to its rated value.

(iii) Find the cost of operating this heater for 50 hours.

*7 A heater is designed to operate at 110 V and 220 V with the same power consumption (Fig c).

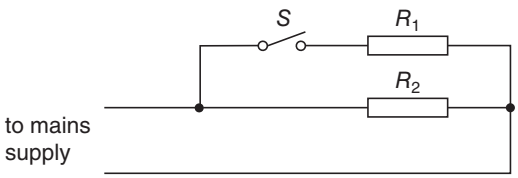


Fig c

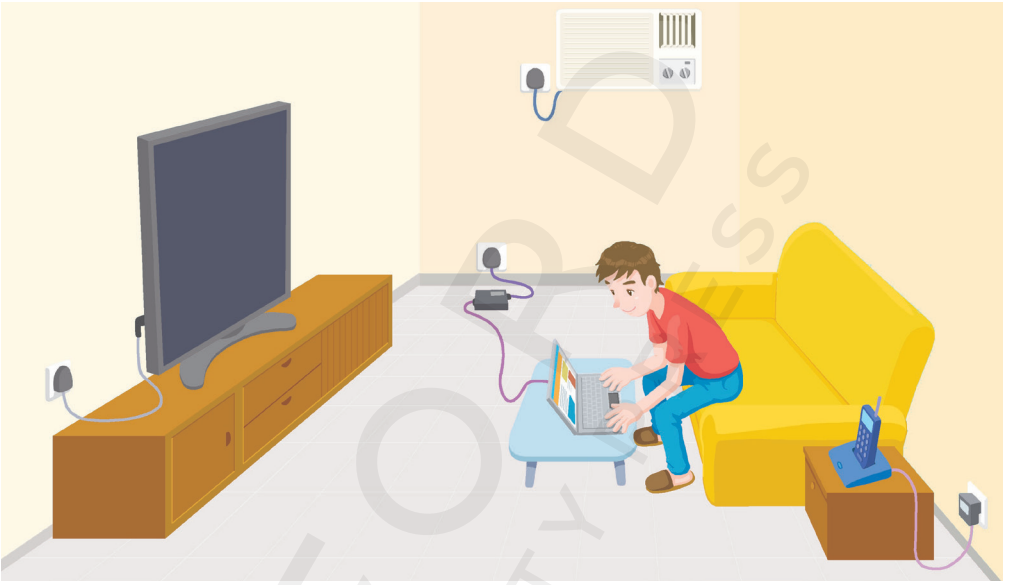
(a) Should switch S be opened or closed when operating at 220 V?

(b) If the resistance of R_1 is 20 Ω,

(i) what is the resistance of R_2?

(ii) what is the rated power of the heater?

4.2 Wiring and safety features of domestic electricity



Every day, we use the electrical energy from the mains to power different kinds of electrical appliances. How are the appliances connected to the mains at the same time and work at their rated values?

DSE goal

Assign the electrical potential of any earthed points as zero.

Understand household wiring.

DSE exam

15(1A)Q28, 20(1A)Q24, 23(1A)Q28

1 Household wiring

a Connecting appliances to mains

Each appliance sold in Hong Kong comes with a three-pin plug. The plug has a live (L), a neutral (N) and an earth (E) pins, each connected to the respective wire of the appliance (Fig 4.2a). A wall socket has three holes for the plug to fit into (Fig 4.2b). An appliance is connected to the mains when its plug is inserted to a wall socket.

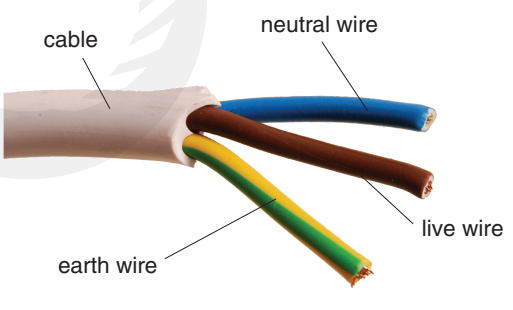


Fig 4.2a *Wires of an appliance.*

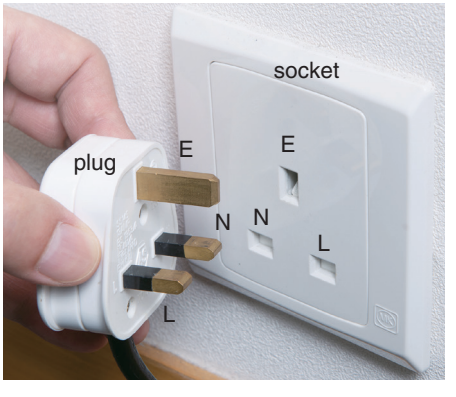


Fig 4.2b *Three-pin plug and socket.*

Know more

Different plugs

Different plug-and-socket systems are in use around the world. The plug used in the Mainland is shown below.



Visit the IEC website to see all types of plugs and sockets.

Useful link

The mains cable entering a house contains two wires, the live and the neutral. Figure 4.2c shows schematically how a socket is connected to the mains cable. Note that:

Like gravitational PE, we can assign the potential of any point in a circuit as the reference point of 0 V. By convention, we assign the earth at 0 V.

1 Current flows only when the live and neutral holes are connected.

2 The neutral wire is connected to the earth, whose electric potential is assigned as 0 V. Therefore, the potential of the neutral wire stays at 0 V.

Cross-link

You learned in Ch 2.2 that a current flows whenever there is a p.d. across a conductor in a closed circuit. It flows from higher potential to lower potential.

We will learn more about a.c. in Ch 4.3.

3 The potential of the live wire varies periodically from a positive peak $+V_p$ to a negative peak $-V_p$. Therefore, the current through an appliance changes direction periodically (Fig 4.2d). Such a current is called an **alternating current** (**a.c.**).

4 The earth wire is a safety design. It connects the metal case of an appliance to the Earth. No current passes through it when the appliance is in normal use. We will learn more about this wire on p.161.

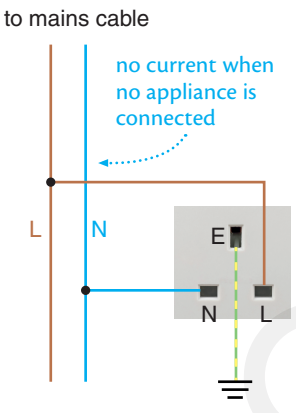


Fig 4.2c *Connection of a socket.*

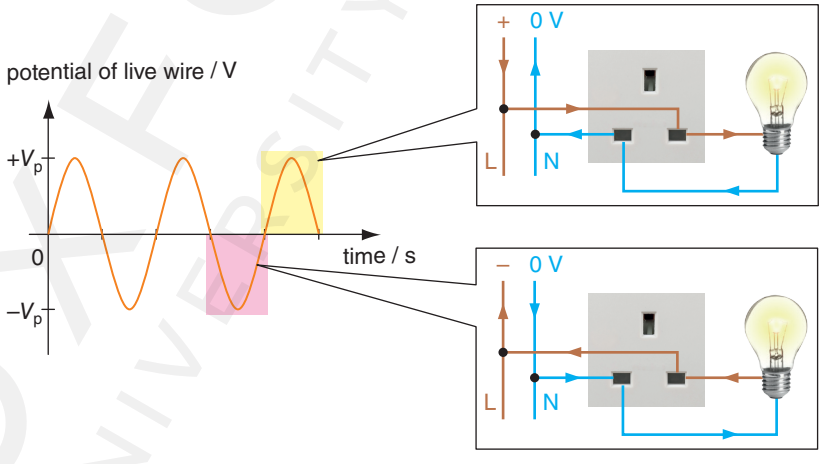


Fig 4.2d *Variation in the potential of the live wire results in the alternate change in the direction of the current.*

The potential of the neutral wire stays at 0 V.
The potential of the live wire keeps changing between a positive peak value and a negative peak value.

Quick check 4

→ E-version in OUP Exercise Platform is also available.

Google Forms Microsoft Forms

True or false:

1 Since the potential of the neutral wire remains 0 V, no current flows in it. (T/F)

2 The potential of the live wire is always higher than that of the neutral wire. (T/F)

3 Current may flow from the neutral wire to the live wire via an appliance. (T/F)

4 Current keeps flowing in the mains wires even if no appliance is connected to the sockets. (T/F)

alternating current (a.c.) 交流電

b Connecting sockets in parallel

All sockets in a house are connected **in parallel** (Fig 4.2e). This connection has the following advantages.

1 Appliances connected to different sockets work **independently**.

2 All sockets work at the **mains voltage**.

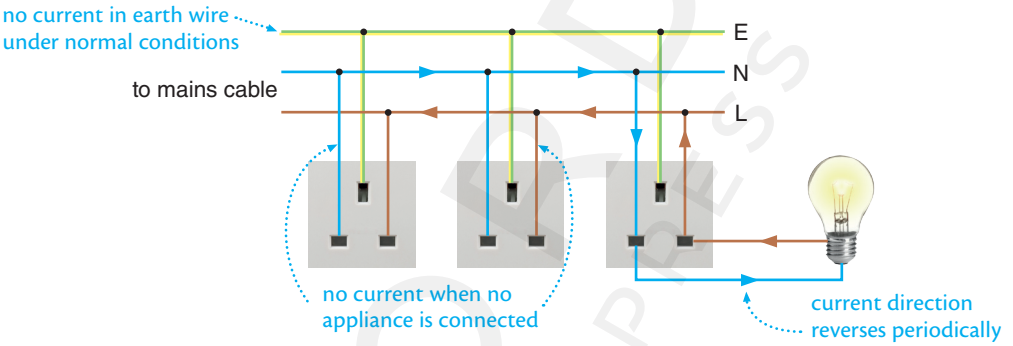


Fig 4.2e *Sockets are connected in parallel.*

c Ring main

Ring mains are primarily used in the UK and Hong Kong.

The wires connecting the sockets may go around a room and form a ring. This is called a **ring main** (Fig 4.2f).

A ring main is also called a ring circuit.

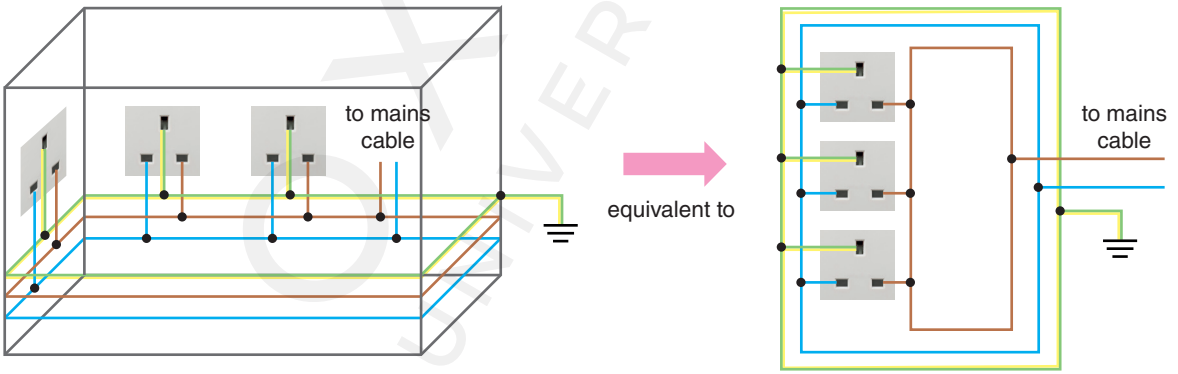


Fig 4.2f *Ring main.*

In the ring main, the live, neutral and earth wires each branches into two paths. Current flows to and from each socket via two separate live and two separate neutral wires. As each wire only carries about half of the total current (Fig 4.2g), thinner and cheaper wires can be used.

Wires that can carry larger current are thicker and more expensive.

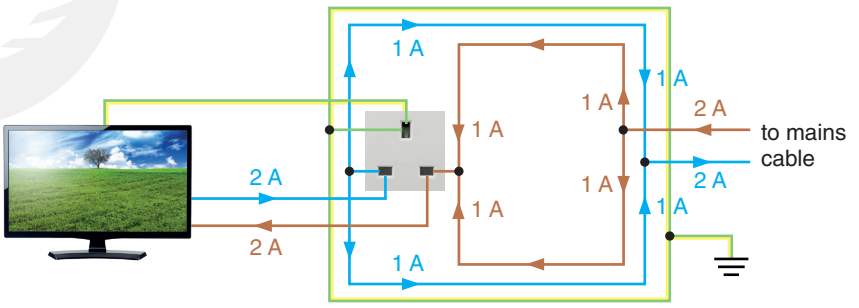


Fig 4.2g *Each wire in a ring main carries about half of the total current.*

Since the load is usually not evenly split across the two branches, the current in each branch may not be the same. In practice, the current rating of the wires in the ring main should be no less than two thirds of the rating of the circuit breaker.

If there is a break in one wire, the ring main can still work as current can flow through the other wire. This can be considered as an advantage of the ring main. However, the remaining wire now needs to carry all the current, and overloading may occur.

ring main 環形幹線

→ Start Oxford Physics XR 'Domestic circuit' and scan this page to play.

Android iOS

The earth terminal in the consumer unit is usually attached to the metal water pipe of the building.

d Circuits around the house

The mains cable enters a house via a **kilowatt-hour meter**. It then branches out in a **consumer unit** into several **parallel** circuits for the ring mains, lighting circuits, air conditioners, etc (Fig 4.2h). This allows all appliances to work at the mains voltage.

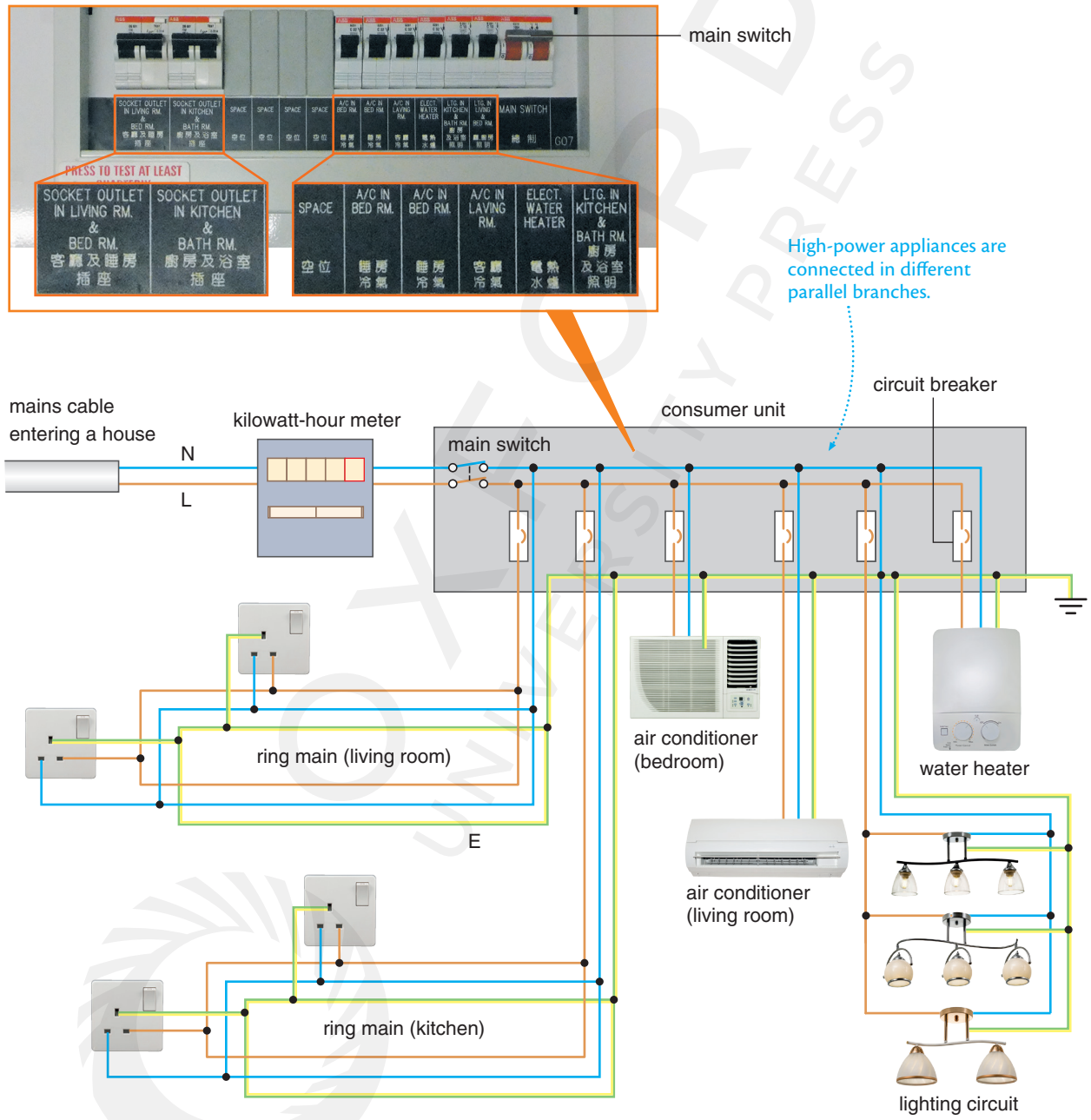


Fig 4.2h *Household wiring plan (showing some branches only).*

Each branch is fitted with a safety device called a **circuit breaker**. We will learn more about it on p.159.

The kilowatt-hour meter records the **electrical energy consumption** of the house. Electricity companies calculate our bills according to the kilowatt-hour meter reading.

kilowatt-hour meter 千瓦時計 consumer unit 供電箱 circuit breaker 斷路器

Google Forms

Microsoft Forms

Quick check 5

→ E-version in OUP Exercise Platform is also available.

1 *True or false:*

(a) Current keeps flowing in the ring main even if no appliance is connected to the sockets. (T/F)

(b) Thinner and cheaper wires can be used in the ring main. (T/F)

(c) All appliances in a house are connected to the ring main. (T/F)

2 What are the advantages of connecting wall sockets in parallel? Put a tick in the corresponding box(es).

☐ The total current drawn by the appliances can be reduced.

☐ The sockets work independently.

☐ The sockets work at the mains voltage.

☐ Thinner and cheaper wires can be used.

DSE goal

Discuss safety aspects of domestic electricity.

Discuss the choice of power cables and fuses based on power rating.

DSE exam

12(1A)Q33, 14(1B)Q8(d), 17(1A)Q25, 18(1B)Q8(b), 19(1B)Q8(a), (b)

2 Safety features

a Fuse

A **fuse** is basically a thin metal wire (Fig 4.2i). It is connected **in series** with the device that it protects. It blows (the metal wire melts) when an excess current flows through it. The circuit becomes open so that the device or the cables will not overheat and catch fire.

Appliances and sockets are commonly protected by fuses.

After the fault causing the excess current is fixed, the blown fuse has to be replaced by a new one.

Flipped classroom

Watch a video and answer the questions.

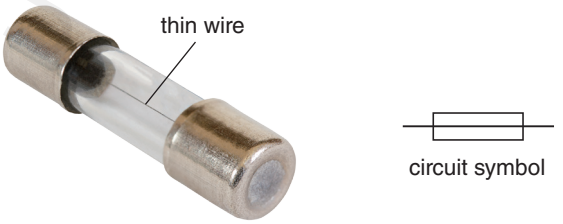


Fig 4.2i A fuse and its circuit symbol.

i Fuse value

Each fuse has a fuse value, or current rating (Fig 4.2j), which is the **maximum** current that the fuse can continuously carry. The fuse in an appliance should have a fuse value **slightly larger** than the normal operating current, which can be worked out from the ratings of the appliance.



Fig 4.2j Fuses of different fuse values.

Video

→ This video demonstrates the result when a large current passes through a fuse.

fuse 保險絲

Example 5 Fuse value

A rice cooker is rated at '220 V, 1000 W'.

(a) Determine which fuse, 3-A, 5-A or 15-A, is suitable for the cooker.

(b) An oven is rated at '220V, 2000 W'. Should the cooker be connected in parallel with the oven to a socket protected by a 13-A fuse?

Solution

(a) By $P = VI$,

$$\text{rated current} = \frac{P}{V}$$

$$= \frac{1000}{220}$$

$$= 4.55 \text{ A}$$

$\therefore$ The 5-A fuse is suitable.

A 15-A fuse allows a current much larger than the rated current. The circuit could overheat and catch fire before the fuse blows in case of a fault.

(b) Rated current of the oven $= \frac{P}{V}$

$$= \frac{2000}{220}$$

$$= 9.09 \text{ A}$$

Total current for the appliances = 4.55 + 9.09 = 13.6 A > 13 A

$\therefore$ The cooker should not be connected in parallel with the oven to the socket. Otherwise, the fuse in the socket will blow.

Appliances for heating usually have high power and draw large current. Do **not** connect them to the same socket.

Google Forms Microsoft Forms

Quick check 6

1 *True or false:*

(a) A fuse should have a fuse value the same as the rated current of the appliance. (T/F)

(b) A fuse is connected in parallel to the device it protects. (T/F)

2 Choose a suitable fuse for the following appliances. All the appliances work at 220 V.

Appliance	Fuse (1 A, 3 A, 5 A, 7 A, 10 A)
180-W TV	
1.5-kW air conditioner	

→ E-version in OUP Exercise Platform is also available.

Learn by practice 4

The following appliances are working at 220 V.

1 Fuses marked 0.3 A, 0.5 A, 3 A and 5 A are available. Which one is most suitable for a 900-W vacuum cleaner?

2 A toaster is fitted with a 6-A fuse. Estimate the maximum value of the operating power of the toaster.

3 Can a 2000-W heater and a 1000-W kettle be connected in parallel to a socket which is protected by a 13-A fuse? Explain.

4 How many 800-W microwave ovens at most can be connected in parallel to a socket which is protected by a 13-A fuse?

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ii Causes of excess current

Example 5 shows a cause of excess current: *overload*. Suppose n appliances are connected in parallel to a socket. The total current I drawn from the socket is the sum of the currents through the appliances (Fig 4.2k). If there are too many appliances, I will be too large, and the socket will be overloaded (Fig 4.2l).

An universal adaptor allows appliances to be connected in parallel to a socket.

A short circuit (Fig 4.2m) is another cause. A short circuit reduces the resistance of the circuit drastically, so the current becomes very large.

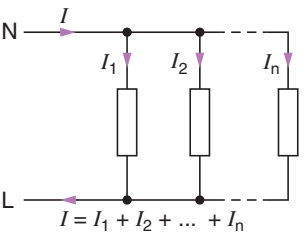


Fig 4.2k *Appliances connected in parallel.*



Fig 4.2l *Overloaded socket.*

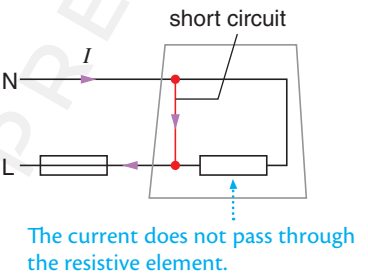


Fig 4.2m *Short circuit.*

Excess current will also result if an appliance operates at a voltage higher than its rated value. For example, if a lamp rated at 110 V is connected to the 220-V mains, the current will be doubled.

Fuses will blow in these three cases.

iii Position of fuse

Current will flow through the person when the potential of the live part is higher or lower than 0 V. Sometimes, we simply say that a point is at a high voltage if it is live.

When a person stands on the ground, his feet are at 0 V. If he touches the live part (potential is not 0 V) of a circuit, there will be a potential difference across his body and a current will flow through him, causing an electric shock. This can be fatal.

A current larger than 0.1 A is likely to be fatal.

A fuse must be placed in the **live wire** (Fig 4.2n). The appliance is no longer live and is safe to touch after the fuse has blown (Fig 4.2o). A switch is placed in the live wire for the same reason.

Simulation

→ This simulation shows the internal structure of the three-pin plug and the socket. It is available on OUP web in PC version (.exe) only.

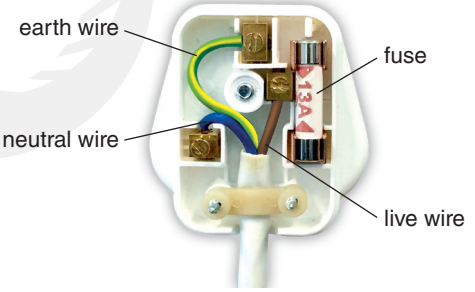


Fig 4.2n *A fuse fitted in the live wire in a plug.*

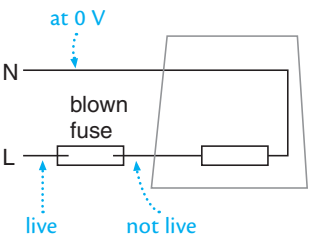


Fig 4.2o *An appliance is not live after the fuse has blown.*

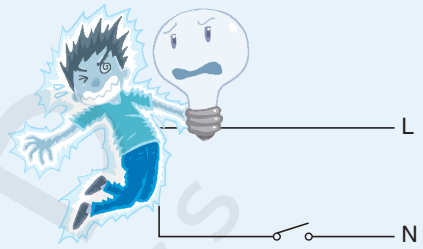
overload 超負荷

Right and wrong

☆ HKDSE 2014 Paper 1B Q8(d)(ii)

✗ The switch of an appliance can be placed in the neutral wire.

✓ The switch must be placed in the live wire. Although the appliance can still work, it is dangerous to place the switch in the neutral wire because the appliance is still live even when the switch is open. This increases the risk of getting an electric shock.



Quick check 7

→ E-version in OUP Exercise Platform is also available.

Google Forms Microsoft Forms

1 *True or false:*

(a) A fuse will never blow if it is placed in the neutral wire. (T/F)

(b) If a person touches the neutral wire of an operating appliance accidentally, the fuse in the appliance will blow. (T/F)

(c) If an appliance operates at a voltage higher than its rated value, its fuse may blow. (T/F)

(d) The current drawn from the mains may increase significantly if there is a short circuit. (T/F)

2 If a switch is placed in the neutral wire, the electrical appliance

A will still work even if the switch is opened.

B will still be live even if the switch is opened.

C will not work even if the switch is closed.

b Circuit breaker

In a consumer unit, the **live wire** in each branch is fitted with a circuit breaker (Fig 4.2p) which functions like a fuse, but in a more convenient way. It is an automatic switch that breaks the circuit if the current exceeds a certain value. It can be reset by simply pushing the switch.

The circuit breakers in different branches have different ratings.

The fault causing excess current must be identified and fixed before resetting the circuit breaker.

Symbol of a circuit breaker:



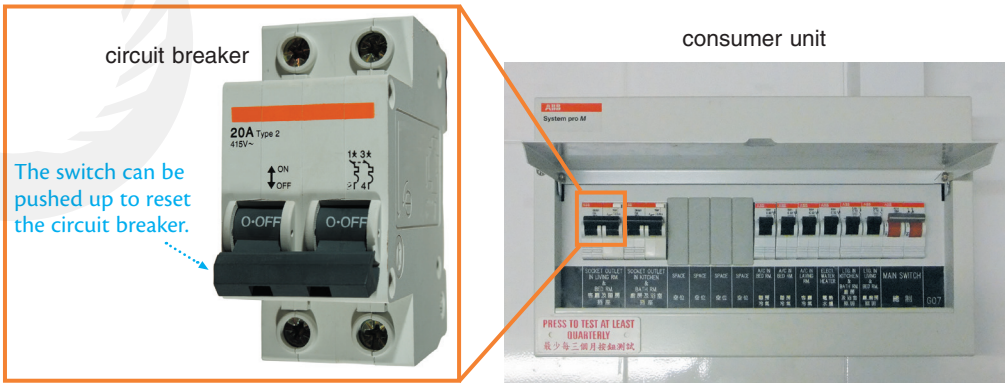


Fig 4.2p *A consumer unit and a circuit breaker.*

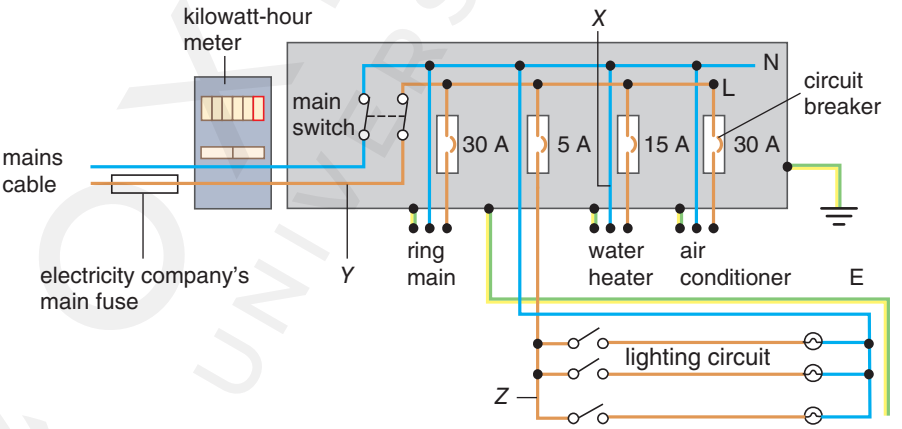
C Branch circuits

Appliances like air conditioners and water heaters draw large current and operate for a long time, so each is connected in a separate branch to avoid overloading. Other advantages of branch circuits include:

1. Each branch circuit operates at the mains voltage.
2. Thinner and cheaper wires can be used in the branch circuits than in the mains cable.
3. The branch circuits can be protected by circuit breakers of the most suitable ratings.
4. Once a circuit breaker is triggered by a fault, only the appliances in that branch would stop. Appliances in other branches can still work. Also, having fewer appliances in a branch allows us to identify the fault more easily.

Example 6 Choice of cables

A simplified household wiring circuit is shown below. The lighting circuit contains three identical lamps connected in parallel.



Which cable, with a current rating of 2 A, 6 A, 17 A, 63 A or 100 A, should be used at X, Y and Z so that the cost of wiring can be minimized?

Tip
The circuit breaker will be triggered if $I > 15$ A.
I is the same in the live and neutral cables.

Tip
The cable must be able to carry the maximum current; otherwise it would burn out before the circuit breaker is triggered.
To minimize the cost, the thinnest (lowest current rating) cable that can carry the current should be used.

Solution

Maximum possible current at X is 15 A.

$\therefore$ A 17-A cable should be used at X.

Maximum possible current at $Y = 30 + 5 + 15 + 30 = 80$ A

$\therefore$ A 100-A cable should be used at Y.

Maximum possible current at $Z = \frac{5}{3} = 1.67$ A

$\therefore$ A 2-A cable should be used at Z.

Quick check 8

→ E-version in OUP Exercise Platform is also available.

Google Forms Microsoft Forms

1 Which of the following statements about the circuits branching out from the consumer unit is correct?

A All branches are fitted with circuit breakers with the same rating.

B All branches work independently.

2 A circuit is fitted with a circuit breaker rated at 30 A. Which cable (rated at 15 A, 28 A and 35 A) is suitable for this circuit?

3 The ratings of the fuse and wires in an appliance are F and W respectively. The rated current of the appliance is I. Arrange F, W and I in ascending order.

A $F < W < I$

B $W < F < I$

C $I < W < F$

D $I < F < W$

d Earth wire

Suppose there is a fault in an appliance such that its live wire gets loose and touches its metal body. The metal body will become live. A person touching the metal body will get an electric shock if the appliance does not have an earth wire (Fig 4.2q).

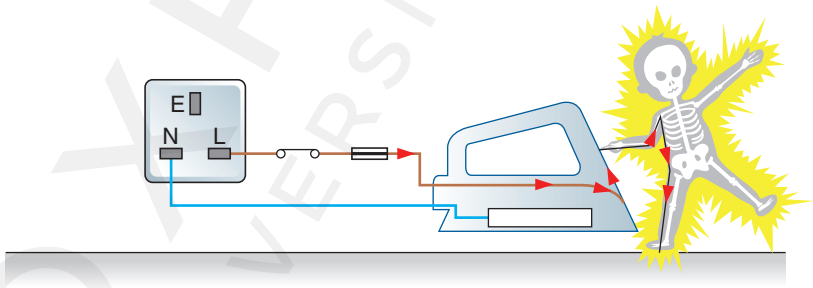


Fig 4.2q *You may get an electric shock due to a fault in an appliance without an earth wire.*

The earth and neutral wires are connected to the earth via separate paths.

▶ In this situation, an earth wire shorting the path from the metal body to the earth can help. A large current will then flow in the live and earth wires and blow the fuse quickly. The metal body is thus prevented from becoming live (Fig 4.2r).

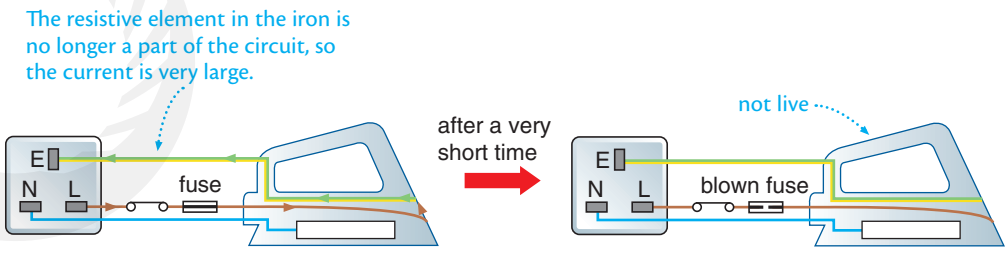


Fig 4.2r *A large current flowing through the earth and live wires will blow the fuse quickly.*

Even if the fuse does not blow, the current through a person touching the metal body is minimal as the earth wire provides a short circuit from the metal body to the earth.

e Double insulation

Appliances with an insulating plastic case do not need an earth wire. They are said to have a double insulation (Fig 4.2s) as they have two layers of insulating material surrounding the live parts: the wire coverings and the plastic case.

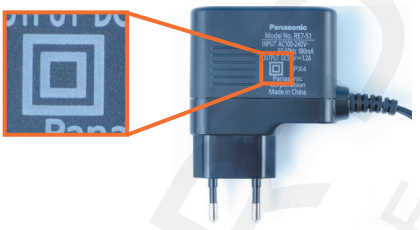


Fig 4.2s Symbol for double insulation.

Safety designs in mains circuit:

1 **Fuse and circuit breaker**
- **break the circuit when current is too large**
- **prevent circuit from overheating**
- **rating slightly larger than normal operating current**
- **installed in live wire**

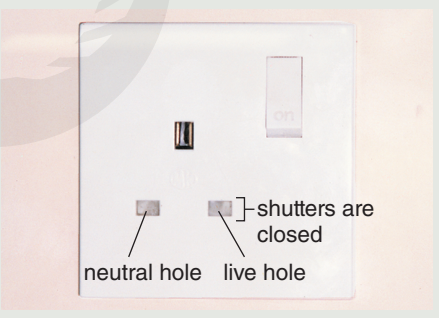
2 **Earth wire**
- **provide a path for current to flow when a fault develops**
- **prevent electric shock**

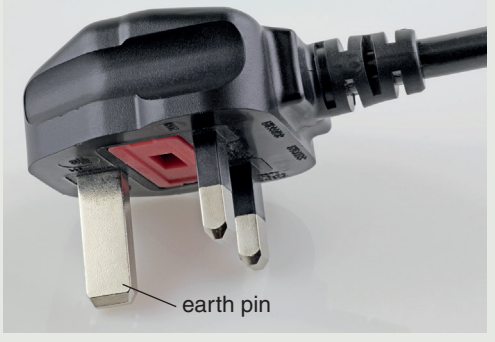
3 **Double insulation (plastic case)**
- **prevent electric shock**

Know more

Shutters on socket

There are shutters covering the live and the neutral holes on a socket. This prevents accidental insertion of conductive objects. The earth pin of a plug is slightly longer than the other two pins to open these shutters. This ensures that the earth wire is connected before the live wire.





The earth pin is thicker than the other two pins so that the plug will not be connected to a socket in the wrong orientation.

May remind Ss that the socket holes are in a rectangular shape, instead of a round shape, so that children cannot put their small fingers into the holes.

Google Forms Microsoft Forms

Quick check 9

→ E-version in OUP Exercise Platform is also available.

1 Which safety design, a fuse, an earth wire or a circuit breaker, do the following statements refer to?

(a) It breaks the circuit by an automatic switch when an abnormally large current flows in the circuit.

(b) It blows when an abnormally large current flows in the circuit.

(c) It earths the metal body of an appliance to prevent an electric shock.

2 *True or false:*

(a) No current passes through the earth wire if an appliance works properly. (T/F)

(b) The current through the neutral wire is smaller than that through the live wire if an appliance works properly. (T/F)

(c) Current will pass through the earth wire if the neutral wire is broken. (T/F)

(d) An appliance with double insulation does not need a fuse. (T/F)

Writing practice

Question 1

Should the switch of an appliance be installed in the neutral wire? Explain. (2 marks)

Answer

No, installing the switch in the neutral wire is dangerous 1A

because the appliance is still live even when the switch is open. 1A

Question 2

State two advantages of connecting wall sockets in parallel instead of in series. (2 marks)

Answer

The sockets can be turned on or off independently. 1A

All sockets can work at the mains voltage. 1A

Practise yourself

1 Should a fuse be installed in the neutral wire? Explain. (2 marks)

2 State two advantages of connecting air conditioners to the mains in parallel instead of in series. (2 marks)

☆ HKDSE 2018 Paper 1B Q8(b)

→ This video demonstrates how this question is solved.

Video

DSE practice 1 | Domestic circuit

A *residual current circuit breaker* (RCCB) (Fig a) is an automatic switch that quickly breaks the circuit whenever there is a small difference between the currents in the live wire L and neutral wire N in the mains cable.

Figure b is a simplified circuit diagram showing how an electric kettle is connected to the mains cable.



Fig a

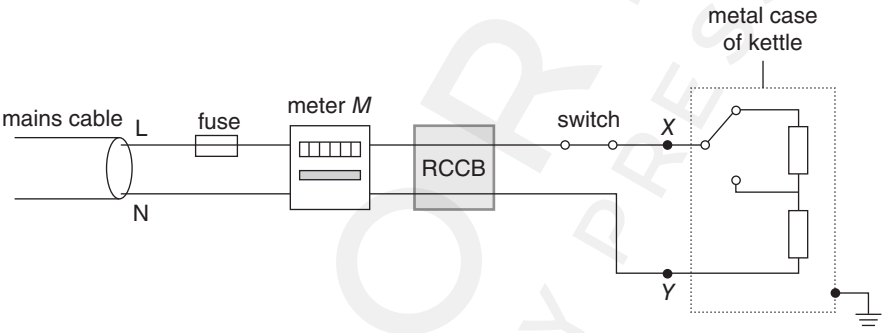


Fig b

Which of the following statements is correct?

A Meter *M* measures the power consumed by the circuit.

B The kettle can still work if a short circuit occurs between point *Y* and the metal case.

C The fuse will blow if a short circuit occurs between points *X* and *Y*.

D The current always flows from point *X* to point *Y* via the kettle.

Solution

M is a kilowatt-hour meter, which measures the energy consumed in the circuit.

∴ A is incorrect.

Common mistake ⚠
Students may wrongly think that the meter measures voltage, current or power.

If a short circuit occurs between point *Y* and the metal case, some current will flow in the earth wire and the current in the neutral wire becomes smaller than that in the live wire. Therefore, the RCCB will break the circuit.

∴ B is incorrect.

Since the current still flows through the resistors in the kettle, its size does not increase and the fuse does not blow.

The mains is an a.c. supply. The current changes direction periodically.

∴ D is incorrect.

If a short circuit occurs between points *X* and *Y*, the resistance of the circuit will decrease drastically and the current will become very large. Therefore, the fuse will blow.

∴ The answer is C.

Before the fuse blows, the currents in the live and neutral wires are the same, so the RCCB will not be triggered.

▶ Revision 4 Q24 (p.182)

residual current circuit breaker 漏電斷路器

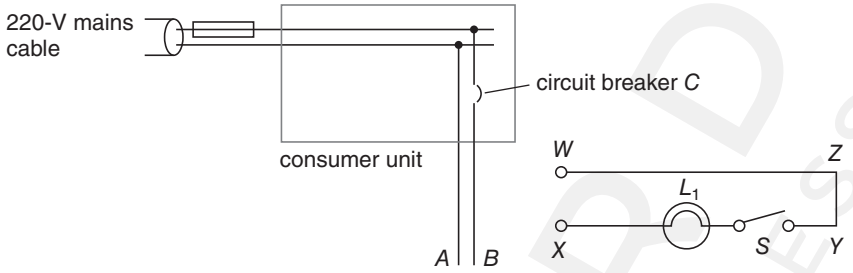
☆ HKDSE 2019 Paper 1B Q8, 2020 Paper 1A Q24

→ This video demonstrates how this question is solved.

Video

DSE practice 2 | Lighting circuit

A simplified household wiring plan is shown below. Wires *A* and *B* branch out at the consumer unit to form the lighting circuit.



(a) Indicate how lamp L_1 should be connected to wires *A* and *B*. (1 mark)

(b) This lighting circuit contains two lamps L_1 and L_2 (not shown in the figure) connected in parallel. Both lamps are rated at '220 V, 100 W'.

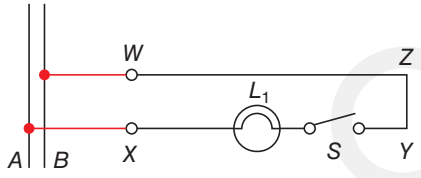
(i) Give one advantage of connecting the lamps in parallel rather than in series. (1 mark)

(ii) Which rating, 0.5 A, 1 A, 3 A and 5 A, is most suitable for circuit breaker *C*? Justify your answer. (3 marks)

(c) One day, lamp L_1 does not light up when switch *S* is closed. A technician finds that points *W*, *X*, *Y* and *Z* are all live. What is wrong with the circuit? (1 mark)

Solution

(a)



Wire *B* with circuit breaker *C* is the live wire. Switch *S* must be connected to the live wire, so *W* is connected to *B*.

1A

(b) (i) Any one of the following: 1A

The lamps work independently.

Both lamps can work at the rated power.

Common mistake

'If one lamp fails due to a **short circuit**, the other lamp can still work' is not correct since *C* would be triggered and both lamps would not work if there is a short circuit.

'Both lamps can work at the **same** voltage' is not the answer since 'same voltage' does not mean the 'rated voltage'.

(ii) By $P = VI$,

$$\text{current through each lamp} = \frac{P}{V}$$

$$= \frac{100}{220}$$

$$= 0.4545 \text{ A} \quad 1M$$

If L_1 or *S* is damaged and breaks the circuit, *X* will not be live. If L_1 is short-circuited, *C* will be triggered. *W*, *X*, *Y* and *Z* will not be live.

Total current through $C = 2 \times 0.4545 = 0.909$ A 1A

∴ The suitable rating is 1 A. 1A

If *X* is disconnected from the neutral wire, no current will flow through L_1.The p.d. across L_1 is 0 and *W*, *X*, *Y* and *Z* are all live.

(c) Point *X* is disconnected from the neutral wire. 1A

▶ Revision 4 Q15 (p.180), Q25 (p.183)

STSE

Safety in using electricity

Electricity is convenient and easy to use, but it can be dangerous if it is not used properly.

Fire

Every year there are fires caused by electricity. These accidents may be caused by:

- overloading a socket (Fig a),
- accidentally short-circuiting an appliance,
- putting flammable materials near the heating parts of an appliance.



Fig a

Electric shock

When we touch somewhere live on a faulty appliance, current flows through our body. If a strong current passes through our body, we will receive an electric shock, which can sometimes be fatal.

The current passing through our body depends on the voltage applied, the resistance of our body, our footwear and the floor covering (Fig b). The resistance of our body when it is wet can be a thousand times less than that when our body is dry.

A current of 1 mA produces a harmless tingling. A larger current can cause burns. A current larger than 0.1 A is likely to be fatal.

If someone gets an electric shock from the mains, the first thing to do is to immediately cut off the electricity supply. First aid can then be administered.

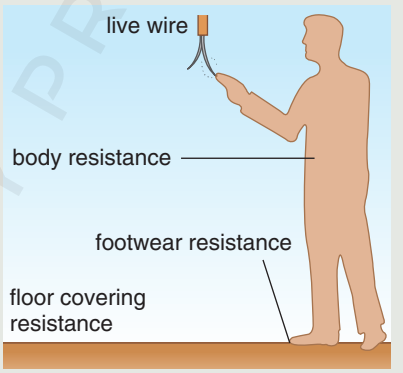
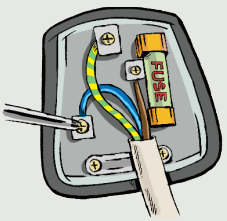


Fig b

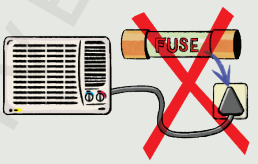
Safety rules



Wire plugs properly and always fit the correct fuse.



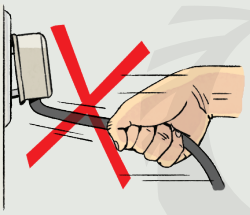
Replace any cables and plugs which are worn or damaged.



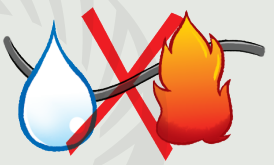
Do not replace a blown-out fuse until the fault is traced and repaired.



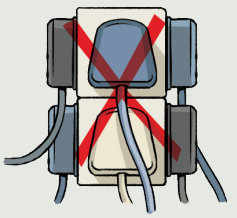
Never switch on/off appliances with wet hands.



Never remove a plug by pulling the cable.



Keep cables away from water and heat sources.



Do not overload a socket.



Keep combustible materials away from lamps and heaters.

More safety tips can be found from the EMSD Household Electrical Safety Handbook on-line.

Useful link

STSE

Saving electricity

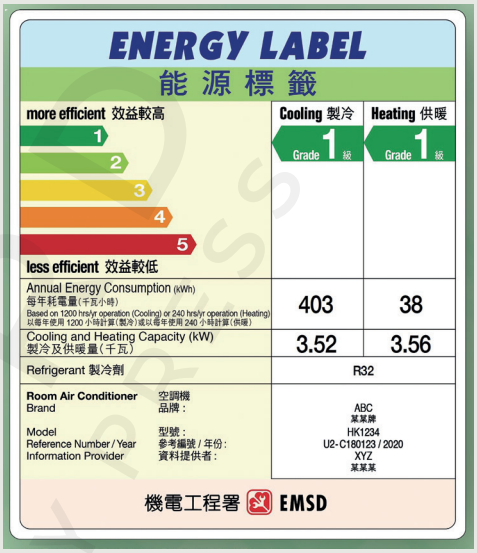
We can save electricity and protect the environment in several means.

1 Use energy-efficient appliances

- Use LED lamps or compact fluorescent lamps instead of filament lamps, which are less energy efficient.
- Some types of appliances are attached with energy labels. We should choose grade 1 products, which are most efficient.

  Visit a website to learn more about energy label.

Useful link



2 Be smart in using electrical appliances

Here are some tips to help us further reduce the everyday use of electricity.

- Avoid using excessive lighting. Turn off the lights when you leave the room.
- Switch off the appliances that are not being used instead of keeping them in stand-by mode.
- Clean the appliances regularly, e.g. the filter of an air conditioner and the coil of a refrigerator can improve their efficiency.
- Keep the doors and windows closed when using an air conditioner and avoid over cooling.

Exercise 4.2

Take the mains voltage as 220 V.

★**1** A 10-A fuse can properly protect a rice cooker. Which of the following can be the operating power of the cooker?

A 500 W
B 1000 W
C 2000 W
D 2500 W

★**2** Which of the following statements about the neutral wire is correct?

A Its potential is always zero.
B The current in it is always zero.
C A fuse is usually installed in this wire.
D It is connected to the metal case of the appliance.

★**3** Which of the following statements about the earth wire in an electrical appliance is/are correct?

(1) The earth wire should be connected to the metal case of the appliance.

(2) The earth wire carries the same current as the live wire when the appliance is working normally.

(3) An electrical appliance with double insulation does not require an earth wire.

A (1) only

B (1) and (3) only

C (2) and (3) only

D (1), (2) and (3)

★**4** A cell phone charger has a plastic case. The earth pin of the plug is made of plastic (Fig a).



Fig a

(a) Why is the 'fake' earth pin needed? (2 mark)

(b) Explain why it is safe to use the charger even without an earth wire. (2 mark)

★**5** (a) Draw lines in Figure b to show how the sockets are connected to the ring main.

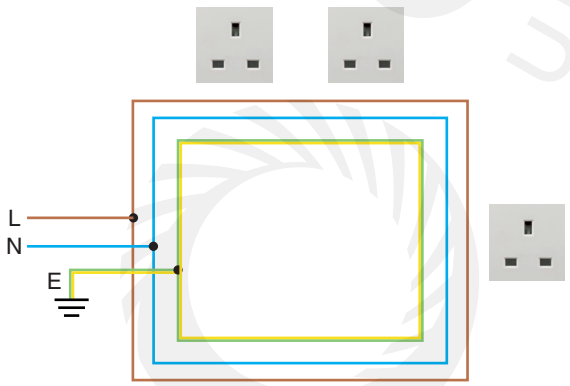


Fig b

(b) State one advantage of connecting the sockets in this way.

(c) Give one advantage of a ring main.

(d) Should an air conditioner be connected to the ring main? Explain.

★**6** A lighting circuit is protected by a 5-A circuit breaker at the consumer unit.

(a) How many 60-W light bulbs can be connected in parallel in the lighting circuit at most?

(b) Wires of rating 0.2 A, 0.3 A, 4 A, 5 A and 6 A are available. What is the current rating of the wires connected directly to

(i) the circuit breaker?

(ii) the light bulbs in (a)?

★**7** Figure c shows two cases when the wire of a toaster gets loose and touches the metal body.

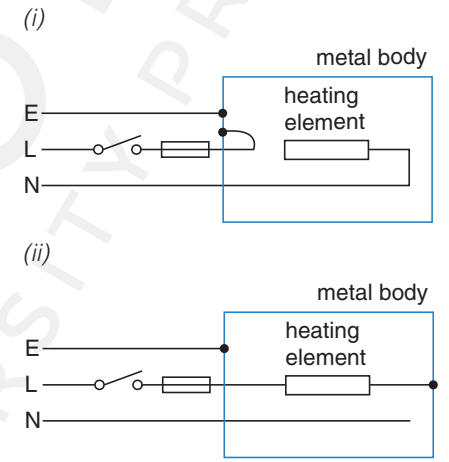


Fig c

In the two cases above,

(i) (1) No (2) Yes (3) No
(ii) (1) No (2) Yes (3) Yes

(a) will current flow through (1) the neutral wire, (2) the earth wire and (3) the heating element just after the switch is closed?

(b) will the fuse blow after the switch is closed for a while?

(c) will a person touching the metal case get an electric shock after the switch is closed for a while?

★**8** Can a hairdryer rated at '110 V, 800 W' and protected by an 8-A fuse work normally if it is connected directly to the 220-V mains? Explain briefly.

★**9** (a) State a function of the earth wire.

(b) Suggest an advantage of keeping the voltage of the neutral wire at 0 V.

(c) Why must the fuse be installed in the live wire?

(d) An appliance with double insulation does not need an earth wire. Explain.

Ext

4.3 Alternating current

We learned on p.146 that the mains voltage in Hong Kong is 220 V. However, we also learned on p.153 that the mains electricity is an alternating current (a.c.) and the potential of the live wire varies from a positive peak $+V_p$ to a negative peak $-V_p$. Is V_p equal to 220 V? Before answering this question, let us learn more about a.c.

DSE goal

Distinguish between d.c. and a.c.

Video

→ This video demonstrates the current and e.m.f. produced by a small a.c. generator.

1 D.c. and a.c.

The current provided by a battery only flows in one direction. This is called a **direct current**, or **d.c.** If we reverse the battery connection, the current will flow in the opposite direction (Fig 4.3a).

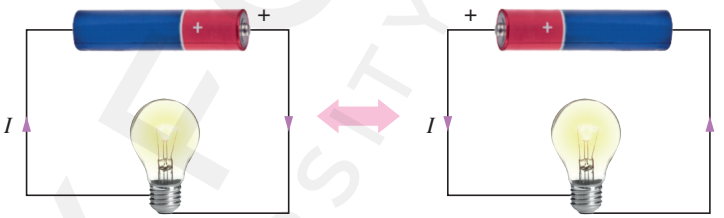


Fig 4.3a *Reversing the battery terminals.*

Suppose we keep reversing the battery terminals such that the current reverses its direction of flow periodically. Such current is an a.c. The bulb keeps lighting up as long as the current flows. A battery is a d.c. power supply while the mains is an a.c. power supply (Fig 4.3b).

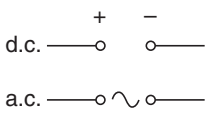


Fig 4.3b *Symbols for power supplies.*

An alternating voltage produces an alternating current. Therefore, an a.c can be described by the current or voltage.

An alternating current (or voltage) changes direction periodically while a direct current does not change direction.

Table 4.3a shows some examples of d.c. and a.c.

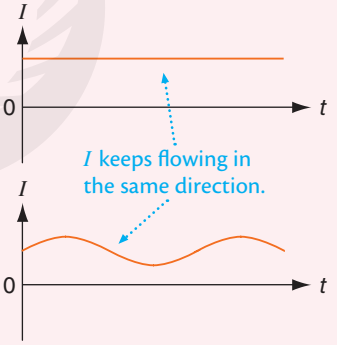
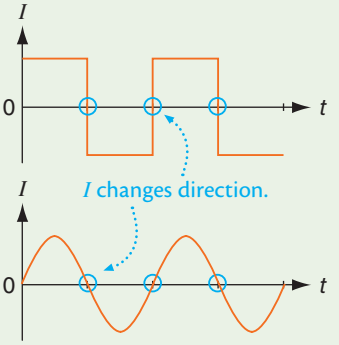
D.c.	A.c.
	

Table 4.3a *Examples of d.c. and a.c.*

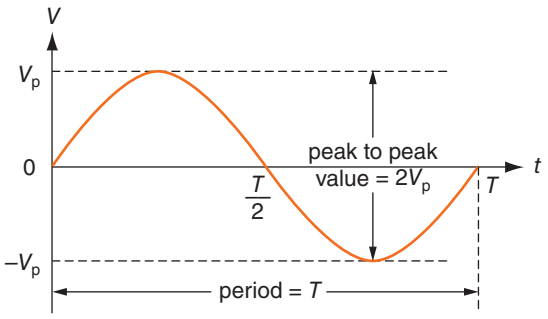
Caution

A d.c. is not necessarily constant.

direct current (d.c.) 直流電

Ext

Figure 4.3c shows some terms describing an a.c. voltage.



period	duration of one cycle (one complete alternation), denoted by T
frequency	number of cycles per second, denoted by $f \left(= \frac{1}{T}\right)$
peak value	maximum value of V, denoted by V_p
peak-to-peak value	difference between positive peak and negative peak, equal to $2V_p$

Fig 4.3c *Terms describing an a.c.*

DSE goal

Define r.m.s. of an a.c. as the steady d.c. which converts electric PE to other forms in a given pure resistance at the same rate as that of the a.c.

2 Root-mean-square value

When a light bulb is connected to an a.c. supply, both the voltage V_{ac} across the bulb and the current I_{ac} through the bulb vary with time. Therefore, the power P_{ac} consumed by the bulb also varies with time. In this case, the average (or mean) power is more meaningful than the instantaneous power.

Suppose the resistance R of the bulb is constant. The value of P_{ac} is given by

$$P_{ac} = I_{ac}^2 R$$

The horizontal bar means taking the mean value, which is a constant.

▶ The mean power $\overline{P_{ac}}$ can be expressed as

$$\overline{P_{ac}} = \overline{I_{ac}^2 R} = \overline{I_{ac}^2} R$$

Suppose the bulb consumes the same power when it is connected to a **steady** d.c. power supply and the current through it is I_{dc}.

$$I_{dc}^2 R = \overline{I_{ac}^2} R$$

$$I_{dc} = \sqrt{\overline{I_{ac}^2}}$$

Cross-link

$\sqrt{\overline{I_{ac}^2}}$ is obtained by taking the square **root** of the **mean** value of the **square** of the current.

You learned root-mean-square speed in Book 1 Ch 5.

Therefore, a steady current I_{dc} of value $\sqrt{\overline{I_{ac}^2}}$ generates the same power as ▶ the a.c. The value $\sqrt{\overline{I_{ac}^2}}$ is called the **root-mean-square (r.m.s.) value** of the a.c., denoted by I_{rms} (Fig 4.3d).

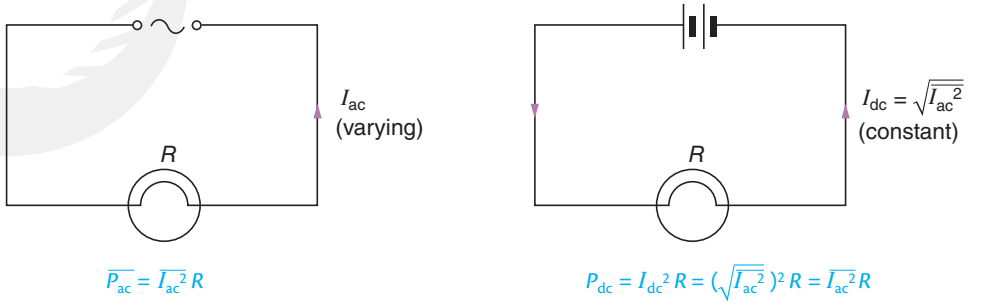


Fig 4.3d *Bulbs of the same power consumption.*

root-mean-square (r.m.s.) value 方均根值

Ext

The r.m.s. value of the voltage can be expressed similarly.

$$I_{rms} = \sqrt{\overline{I_{ac}^2}}$$

$$V_{rms} = \sqrt{\overline{V_{ac}^2}}$$

The mean power $\overline{P}$ consumed by a load connected in an a.c. circuit can be expressed in terms of the r.m.s. values as follows:

$$\overline{P} = I_{rms}^2 R = \frac{V_{rms}^2}{R} = V_{rms} I_{rms}$$

Know more

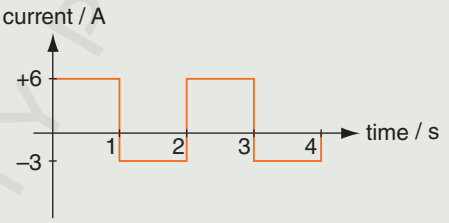
An example showing how r.m.s. value is calculated

For the a.c. shown in the figure,

$$I_{rms} = \sqrt{\overline{I^2}}$$

$$= \sqrt{\frac{6^2(1) + (-3)^2(1)}{2}}$$

$$= 4.74 \text{ A}$$



DSE goal

Relate r.m.s. and peak values of a sinusoidal a.c.

DSE exam

14(1A)Q30, 16(1A)Q30, 17(1A)Q29, 19(1A)Q30, 21(1A)Q28, 22(1A)Q30

3 Sinusoidal a.c.

The mains electricity is a **sinusoidal** a.c. V and I vary with time t in the form of a sine curve (Fig 4.3e).

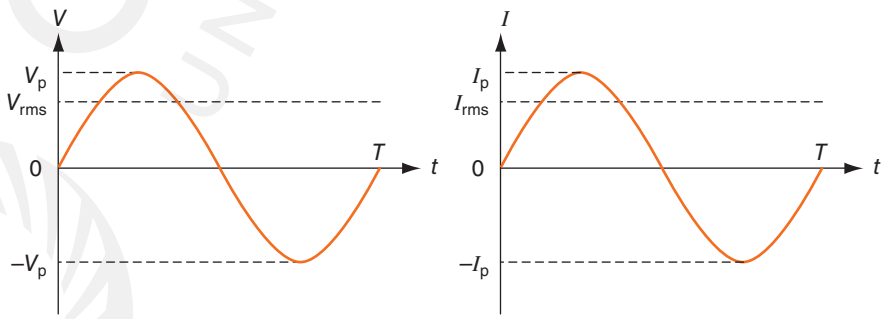


Fig 4.3e *Sinusoidal a.c.* May tell Ss that $V = V_p \sin\frac{2\pi t}{T}$.

Know more

Sinusoidal a.c.

We can derive $I_{rms} = \frac{I_p}{\sqrt{2}}$ for sinusoidal a.c.

▸ p.184

Caution

These relationships only apply to sinusoidal a.c. ▸

It can be shown that

r.m.s. **voltage** (V) ⋯ $V_{rms} = \frac{V_p}{\sqrt{2}}$ ⋯ peak **voltage** (V)

r.m.s. **current** (A) ⋯ $I_{rms} = \frac{I_p}{\sqrt{2}}$ ⋯ peak **current** (A)

Ext

In Hong Kong, the mains electricity has a r.m.s. voltage of 220 V and a frequency of 50 Hz. By $V_{rms} = \frac{V_p}{\sqrt{2}}$, the peak voltage is

The mains current changes direction 100 times each second.

$$V_p = \sqrt{2}\,V_{rms} = \sqrt{2} \times 220 = 311 \text{ V}$$

Figure 4.3f shows how the mains voltage varies with time.

$T = \frac{1}{f} = \frac{1}{50} = 0.02$ s

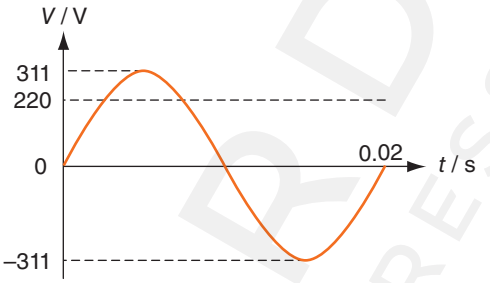


Fig 4.3f *The mains voltage in Hong Kong.*

Note that:

1 The r.m.s. value **does not depend on the period or frequency**. In some countries, the mains electricity has a frequency of 60 Hz. The relationship $V_{rms} = \frac{V_p}{\sqrt{2}}$ is still valid.

2 The rated power of an a.c. appliance is the mean power, and the rated voltage and current are r.m.s values.

Example 7 R.m.s. value and peak value

An oven is rated at '220 V, 1500 W'. It is connected to the mains which has a r.m.s. voltage of 220 V and a frequency of 50 Hz.

(a) Find the resistance R of the oven.

(b) Find the maximum instantaneous current (i.e. peak current) I_p through the oven.

(c) If the oven is connected to the mains in another city, which has a voltage of 120 V and a frequency of 60 Hz, what is the maximum instantaneous power P_p consumed by the oven?

Solution

(a) $R = \frac{V_{rms}^2}{\overline{P}} = \frac{220^2}{1500} = 32.2667\ \Omega \approx 32.3\ \Omega$

(b) $I_{rms} = \frac{\overline{P}}{V_{rms}} = \frac{1500}{220} = 6.82$ A

$I_p = \sqrt{2}\,I_{rms} = \sqrt{2} \times 6.82 = 9.64$ A

(c) $V_p = \sqrt{2}\,V_{rms} = \sqrt{2} \times 120 = 169.7$ V

$P_p = \frac{V_p^2}{R} = \frac{169.7^2}{32.2667} = 893$ W

Caution

$P_p \neq \sqrt{2}\,\overline{P}$

$P_p = \frac{V_p^2}{R}$

$= \frac{(\sqrt{2}\,V_{rms})^2}{R}$

$= 2 \times \frac{V_{rms}^2}{R}$

$= 2\,\overline{P}$

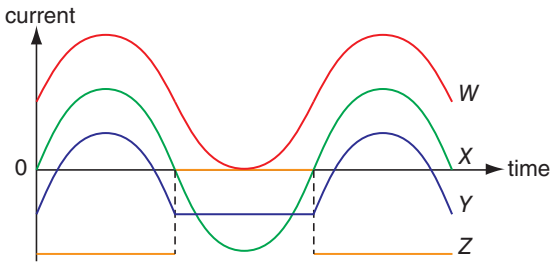
Ext

Quick check 10

E-version in OUP Exercise Platform is also available.

Google Forms Microsoft Forms

1 Determine whether each of the waveforms below shows an a.c. or a d.c.



2 *True or false:*

(a) When a light bulb is connected to an a.c. power supply, the mean power consumed by it varies with time. (T/F)

(b) The e.m.f. of a d.c. power supply must be constant. (T/F)

(c) The peak voltage of an a.c. is always equal to $\sqrt{2}\,V_{rms}$. (T/F)

(d) If the frequency of an a.c. increases and the peak voltage remains unchanged, the r.m.s. voltage will increase. (T/F)

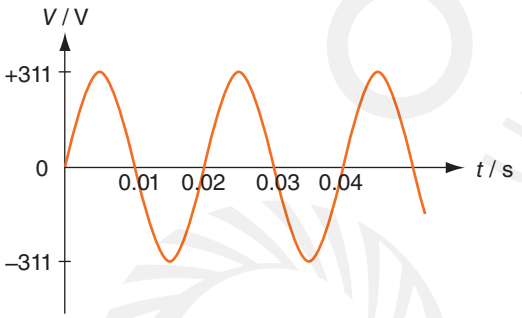
3 Complete the following table describing a sinusoidal a.c.

	I_{rms} / A	I_p / A
(a)	5	
(b)		2

Learn by Practice 5

1 State the main difference between an a.c. and a d.c.

2 The following figure shows how the output voltage V of the mains varies with time t.



A light bulb of resistance 800 Ω is connected to the mains. Find

(a) the frequency of the voltage,

(b) the r.m.s. voltage across the bulb,

(c) the peak-to-peak voltage across the bulb,

(d) the peak current through the bulb,

(e) the peak-to-peak current through the bulb,

(f) the average power consumed by the bulb,

(g) the peak power consumed by the bulb.

3 Consider a lamp connected to a sinusoidal a.c. power supply. The peak voltage is V_p and the peak current is I_p. The lamp has a resistance R and an average power $\overline{P}$.

(a) If I_p = 15 A, find I_{rms}.

(b) If V_{rms} = 110 V, find V_p.

(c) If V_{rms} = 220 V and $\overline{P}$ = 100 W, find I_{rms} and R.

(d) If V_p = 340 V and I_{rms} = 5 A, find R and $\overline{P}$.

(e) If $\overline{P}$ = 80 W and R = 5 Ω, find V_p, I_p, V_{rms} and I_{rms}.

4 A lamp rated at '220 V, 100 W' is connected to the mains (220 V, 50 Hz). Find

(a) the r.m.s. current I_{rms},

(b) the peak current I_p,

(c) the electrical energy consumed by the lamp in 10 minutes.

5 When an electric iron of resistance 12.1 Ω is connected to the mains in country A, the maximum instantaneous power that it consumes is 2000 W. Find the mains voltage.

Ext

Let us verify the relationship between V_{rms} and V_p of a sinusoidal a.c. in Experiment 4a.

Video

→ This video demonstrates Expt 4a.

✎ Switch on the ray box for at least 5 minutes before taking the initial joulemeter reading and starting to time 5 minutes.

Do not touch the hot ray-box lamp.

Experiment 4a / R.m.s. and peak values

1 Connect a ray box to a power supply via a joulemeter. Set the output voltage to 12 V sinusoidal a.c.

2 Measure the peak-to-peak voltage across the ray box by a CRO.

3 Measure the electrical energy consumed by the ray box in 5 minutes.



4 Measure the resistance R of the ray box by using a multimeter, immediately after the power supply is switched off.

Discussion

✎ Q1–3 depend on experimental data.

1 What is the peak voltage V_p across the ray box?

2 What is the mean power consumed by the ray box?

3 What is V_{rms} across the ray box as calculated from the mean power consumption?

4 Is the calculated value of V_{rms} equal to $\frac{V_p}{\sqrt{2}}$?

Skill — Experiment

→ Teaching notes and exercises are available on OUP web.

Y-gain and time base of CRO

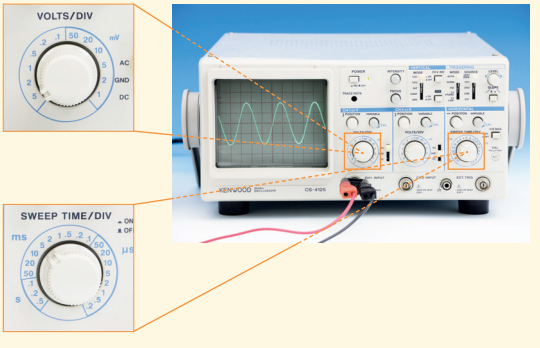
To measure an alternating e.m.f. with a CRO, the vertical and horizontal scales are determined by the settings of the Y-gain and the time base respectively.

Y-gain

The scale is usually in volts per division (V / div), i.e. how many volts are represented by 1 division.

Time base

The scale is usually in seconds per division (s / div), i.e. how many seconds are represented by 1 division.



Ext

☆ HKDSE 2014 Paper 1A Q30, 2016 Paper 1A Q30, 2019 Paper 1A Q30

→ This video demonstrates how this question is solved.

Video

DSE practice 3 | Light bulb connected to d.c. and a.c.

A filament bulb is connected to a power supply. When the power supply provides steady d.c. of voltage V_1, the power consumed by the bulb is P. When the power supply provides sinusoidal a.c. of r.m.s. voltage V_2, the power consumed by the bulb becomes $2P$. Assume the resistance of the bulb is constant.



(a) Express V_2 in terms of V_1. (2 marks)

(b) Then the peak voltage of the a.c. power supply is doubled and the frequency is halved. Express the new r.m.s. voltage V_3 in terms of V_2. (2 marks)

(c) When the voltage of the power supply remains unchanged, how will the power consumed change if the bulb is replaced by another one with a higher resistance? (1 mark)

Solution

(a) When d.c. is used,

$$P = \frac{V_1^2}{R}$$ 1M

When a.c. is used,

$$2P = \frac{V_2^2}{R}$$

$$2 \times \frac{V_1^2}{R} = \frac{V_2^2}{R}$$

$$V_2 = \sqrt{2}V_1$$ 1A

(b) $V_{rms} = \frac{V_p}{\sqrt{2}} \propto V_p$ 1M

$\therefore$ V_p doubled $\Rightarrow$ V_{rms} doubled

$\therefore$ $V_3 = 2V_2$ 1A

(c) Decrease 1A

Common mistake
Some students cannot relate V_p and V_{rms}.

Common mistake
Some students wrongly think that the power consumed will increase when the resistance increases.
By $P = \frac{V^2}{R}$, for the same V,
$R \uparrow \Rightarrow P \downarrow$

▶ Revision 4 Q13, 14 (p.179)

Ext

Exercise 4.3

(For Q1–2.) A heater of resistance R is connected to the mains which gives the sinusoidal voltage as shown (Fig a).

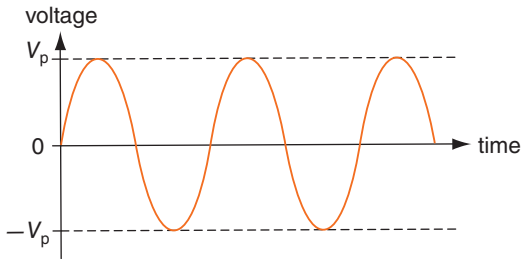


Fig a

1 Find the r.m.s. current passing through the heater.

A $\frac{V_p}{R}$

B $\frac{V_p}{\sqrt{2}R}$

C $\frac{V_p}{2R}$

D $\frac{V_p}{4R}$

★**2** Which of the following expressions give(s) the average power consumed by the heater?

(1) $\frac{V_{rms}^2}{R}$

(2) $\frac{V_p^2}{R}$

(3) $\frac{V_p^2}{2R}$

A (1) only B (3) only

C (1) and (2) only D (1) and (3) only

★**3** Which of the following statements is/are correct?

(1) A d.c. may have varying magnitude.

(2) An a.c. must reverse its direction periodically.

(3) The peak voltage from the mains in Hong Kong is 220 V.

A (2) only B (1) and (2) only

C (1) and (3) only D (2) and (3) only

★**4** A '220 V, 100 W' light bulb is connected to a power supply providing sinusoidal a.c. with a peak voltage equal to 220 V. What is the average power consumed by the bulb?

A 50 W

B 70.7 W

C 100 W

D 141 W

★**5** A kettle is powered by a sinusoidal a.c. Its heating element has a resistance of 48 Ω and its average power is 2400 W.

(a) What is the peak voltage across the heating element?

(b) How will its power change if

(i) the peak voltage of the a.c. doubles;

(ii) the frequency of the a.c. doubles?

★**6**

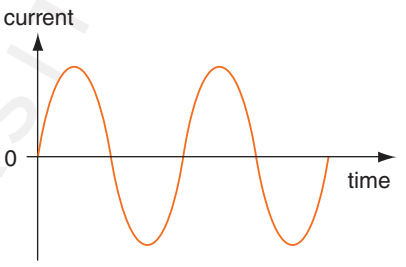


Fig b



Fig c

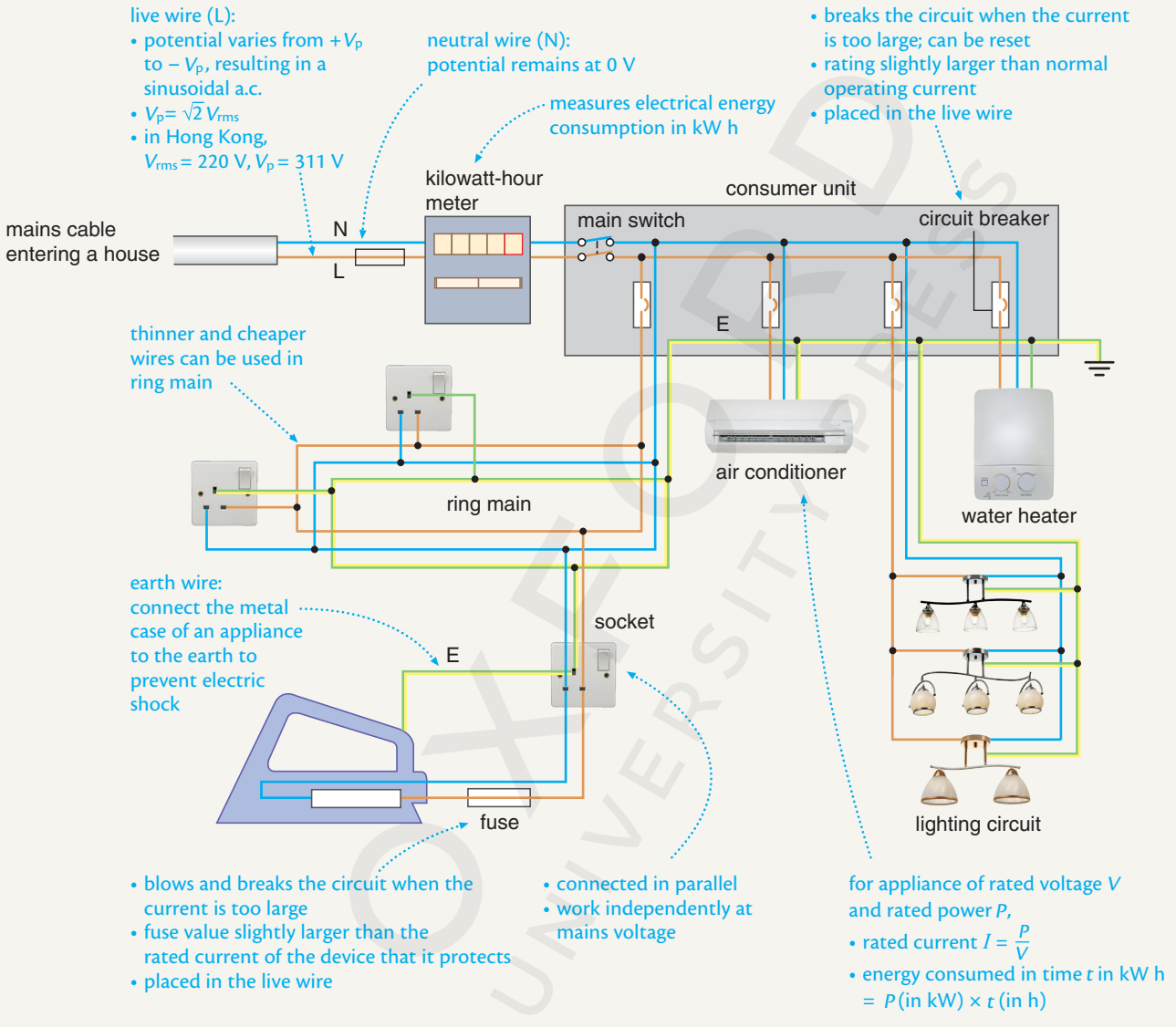
The above figures show sinusoidal current (Fig b) and steady current (Fig c) from electric sources.

(a) Determine whether the current shown in each figure is an a.c. or a d.c. Explain your answer. Fig b: a.c.; Fig c: d.c.

(b) The two currents are passed in turn through a resistor, and the average power dissipated by the resistor is the same in both cases. Given that I_0 = 5 A, find the peak value of the sinusoidal current.

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Key concepts 4



Key terms

1 alternating current (a.c.) 交流電 p.153
2 circuit breaker 斷路器 p.155
3 consumer unit 供電箱 p.155
4 direct current (d.c.) 直流電 p.169
5 fuse 保險絲 p.156
6 kilowatt-hour (kW h) 千瓦小時 p.149
7 kilowatt-hour meter 千瓦時計 p.155
8 mains supply 市電電源 p.146
9 ring main 環形幹線 p.154
10 root-mean-square (r.m.s.) value 方均根值 p.170

Revision 4

Unless otherwise specified, take the mains voltage as 220 V and the cost of electricity as $1.1 per kW h.

Multiple-choice questions

(For Q1–2.) An electrical appliance is faulty, and the live wire accidentally touches the metal case (Fig a).

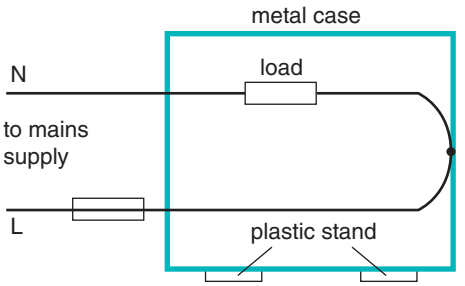


Fig a

★1 4.2 If no earth wire is installed, which of the following statements is/are correct?

(1) The fuse will blow.
(2) The metal case will remain live.
(3) The appliance will not work.

A (1) only B (2) only
C (1) and (3) only D (2) and (3) only

★2 4.2 If an earth wire is installed, which of the following statements is/are correct?

(1) The fuse will blow.
(2) The metal case will remain live.
(3) The appliance will not work.

A (1) only B (2) only
C (1) and (3) only D (2) and (3) only

★3 4.2 A kettle is plugged into a wall socket and works normally. Which of the following statements is/are correct?

(1) The current flows through the kettle in alternate directions.
(2) The live, neutral and earth wires form a closed circuit for current to flow through.
(3) The current in the live wire is as large as that in the neutral wire.

A (1) only B (1) and (3) only
C (2) and (3) only D (1), (2) and (3)

★4 4.2 A fan is connected to the 220-V mains and works normally. The current in its live wire is 0.25 A. Which of the following correctly shows the current I and potential V of its neutral wire and earth wire?

	Neutral wire		Earth wire	
	I / A	V / V	I / A	V / V
A	0	0	0	0
B	0.25	0	0	0
C	0.25	–220	0	0
D	0.125	220	0.125	220

★5 4.1 The ratings of bulb X and Y are '110 V, 25 W' and '110 V, 50 W' respectively. The two bulbs are connected in series to a 220-V mains supply. Which of the following statements is correct?

A The power consumption of X is half the power consumption of Y.
B The power consumption of X is double the power consumption of Y.
C X and Y are both working at their rated conditions.
D The total power consumption is 75 W.

★6 4.2 Which of the following is/are the function(s) of the earth wire?

(1) It prevents the appliance from overheating.
(2) It prevents users from getting an electric shock.
(3) It keeps the potential of the neutral wire at 0 V.

A (1) only B (2) only
C (1) and (3) only D (2) and (3) only

★7 4.2 Which of the following is/are the advantage(s) of using several parallel branch circuits over one big circuit to supply electricity to all appliances in a house?

(1) Cheaper cables can be used in the branches.
(2) Appliances and circuits can be protected by suitable circuit breakers.
(3) A fault usually affects one branch but not the whole house.

A (1) only B (1) and (3) only
C (2) and (3) only D (1), (2) and (3)

*8 Ext 4.3 A bulb rated '220 V, 60 W' is connected to a 220-V mains supply. Which of the following statements is/are correct?

(1) Its highest instantaneous power is 60 W.

(2) Its average power is 60 W.

(3) It has the same output power if it is connected to a 220-V d.c. power supply.

A (1) only B (2) only

C (1) and (3) only D (2) and (3) only

*9 Ext 4.3 A heater consumes power P when connected to an a.c. supply with a peak voltage V_0. What is the power consumed by the heater if it is connected to a steady d.c. supply with voltage $2V_0$?

A $\sqrt{2}\,P$ B $2P$

C $4P$ D $8P$

*10 Ext 4.3 An electrical appliance is designed to operate at the mains voltage as shown (Fig b). Its rated power is 1000 W.

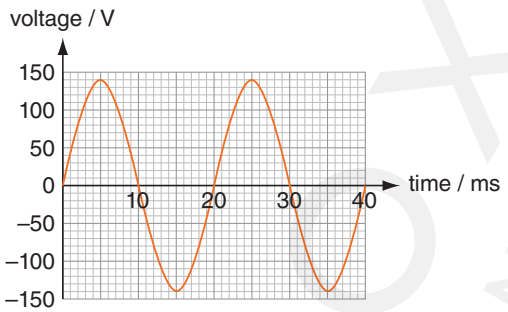


Fig b

Which fuse value is the most suitable for the appliance?

A 3 A B 5 A

C 10 A D 12 A

**11 4.2 Figure c shows the circuit of an electric heater.

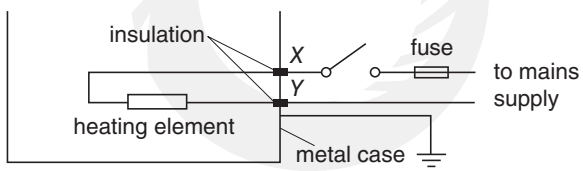


Fig c

The fuse blows when the heater is switched on. Which of the following could be the cause?

(1) The insulation at point X is worn out such that the wire touches the metal case.

(2) The insulation at point Y is worn out such that the wire touches the metal case.

(3) The heating element is broken.

A (1) only B (3) only

C (1) and (2) only D (2) and (3) only

Refer p.161

**12 4.1 An electric stewing pot has three heating elements, each of resistance 100 Ω (Fig d).

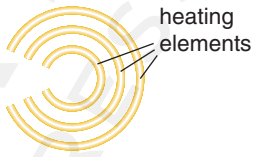
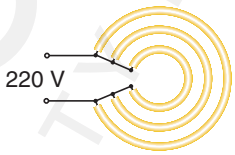


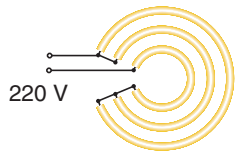
Fig d

Which of the following connections can boil water the fastest?

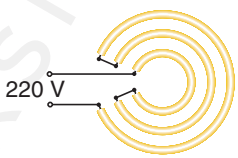
A



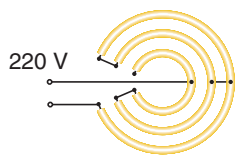
B



C



D



Refer p.148

13 HKDSE 2016 Paper 1A Q30

Ext 4.3 A sinusoidal a.c. of a certain frequency delivers a r.m.s. voltage V_{rms}. If its frequency is doubled and its peak voltage is halved, what would be the r.m.s. voltage?

A $\frac{1}{2}V_{rms}$ B $\frac{1}{\sqrt{2}}V_{rms}$

C $\frac{1}{2\sqrt{2}}V_{rms}$ D V_{rms}

14 HKDSE 2019 Paper 1A Q30

Ext 4.3 The power consumption of the heating element of an electric heater connected to an a.c. mains can be increased by

(1) increasing the electrical resistance of the heating element.

(2) increasing the frequency of the a.c. voltage.

(3) increasing the r.m.s. value of the a.c. voltage.

A (1) only B (3) only

C (1) and (2) only D (2) and (3) only

15 4.2 **HKDSE 2020 Paper 1A Q24**

Figure e shows part of a domestic lighting circuit in which the bulb *L* does not light up when switch *S* is closed.

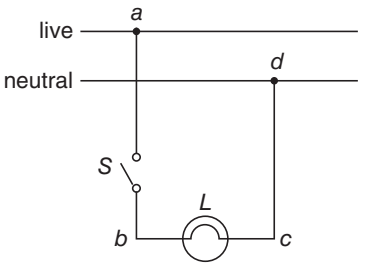


Fig e

The circuit is then checked with switch *S* closed. Using a voltage tester to touch points *b* and *c* in turns, the tester indicates that both points are at high voltage. When touching points *a* and *d* in turns, the tester indicates only point *a* is at high voltage. Which of the following can be a reason for the fault?

A The switch *S* has been damaged.

B The filament of bulb *L* has been burnt out and becomes an open circuit.

C There is a short circuit between *a* and *d*.

D There is an open circuit between *c* and *d*.

Conventional questions

★**16** 4.2 The wire in an iron gets loose and touches the metal body as shown (Fig f).

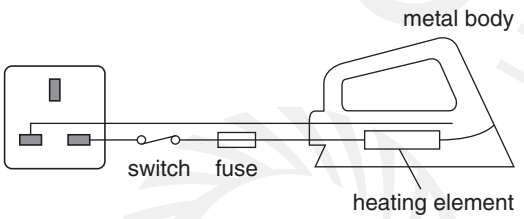


Fig f

(a) Suppose the iron does not have an earth wire. What would happen if a person touches the metal body? (1 mark)

(b) Suppose the iron has an earth wire.

(i) Explain how the earth wire can prevent an electric shock. (2 marks)

(ii) Will the fuse blow? Explain. (2 marks)

(c) Does a hair dryer with double insulation need an earth wire? Explain. (2 marks)

★**17** Misc A 10-A fuse is at 25 °C initially. The fuse wire melts after a current of 20 A flows through it for time *T*. Assume that the resistance of the wire remains at 0.008 Ω.

(a) Explain the function of a fuse. (2 marks)

(b) Estimate the power dissipated by the fuse wire when the 20-A current is flowing through it. (2 marks)

(c) The fuse wire melts when it is heated to 700 °C. Its heat capacity is 0.03 J $°C^{-1}$. Estimate the value of *T*. (3 marks)

(d) State an assumption that you made in (c). Explain how this assumption affects the answer in (c). (3 marks)

★**18** 4.2 Figure g shows a part of a household electrical wiring plan. At the consumer unit, the wires branch out into several parallel circuits. The branch leading to a ring main is fitted with a 30-A circuit breaker.

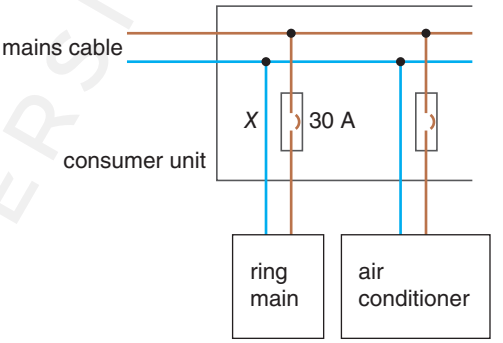


Fig g

(a) Should the current rating of wire *X* be larger or smaller than 30 A? Explain. (2 marks)

(b) Why is the air conditioner not connected to the ring main? (2 marks)

(c) Five sockets, each protected by a 13-A fuse, are connected to the ring main.

(i) State a difference between a circuit breaker and a fuse. (1 mark)

(ii) Are the sockets connected in parallel or in series? Suggest two advantages of this kind of connection. (3 marks)

(iii) Jimmy says that the circuit breaker will certainly be triggered because the total current drawn by the sockets is greater than 30 A. Comment on this statement. (2 marks)

(iv) Karen says that the circuit breaker will never be triggered because its rating is larger than the rating of the fuses protecting the sockets. Comment on this statement. (2 marks)

*19 4.2 Read the following passage about a **two-way switch** and answer the questions that follow.

A lamp is usually controlled by a switch. Sometimes, two switches are installed to a lamp for convenience. The two switches are installed at different positions and the lamp can be turned on and off by either switch. The two switches form a two-way switch. This is especially useful for lamps in a long corridor or staircase (Fig h).



Fig h

The two lamps in Figure h are controlled by a two-way switch. They can be turned on and off together at the bottom or at the top of the staircase. They both operate at the mains voltage of 220 V.

(a) Complete Figure i to show how the lamps are connected to the live (L) wire, the neutral (N) wire and the two-way switch. (3 marks)

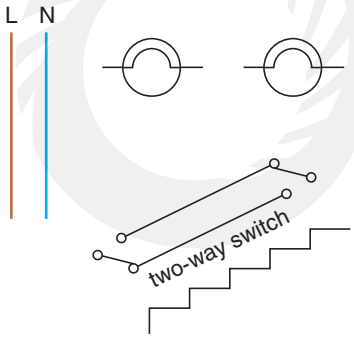


Fig i

(b) The resistance of each lamp is 605 Ω. Find the cost of operating the two lamps for 60 hours. (2 marks)

*20 Misc Figure j shows schematically how a 2000-W water heater is connected to a 220-V socket.

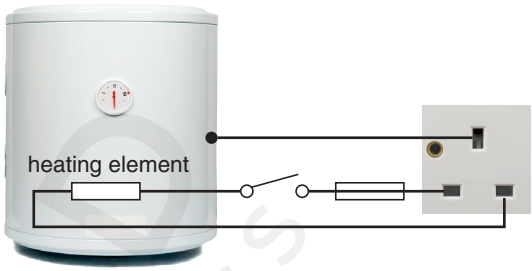


Fig j

(a) The circuit is wrongly wired. Explain. (2 marks)

(b) State a function of the earth wire. (1 mark)

(c) Ext Find the r.m.s. and peak values of the current through the heater. (4 marks)

(d) Which fuse, 2-A, 9-A, 10-A or 13-A, is suitable for the heater? Explain. (2 marks)

(e) The water heater is connected to the mains in a separate circuit parallel to the ring main. Why is it not connected to the ring main? (2 marks)

(f) The heater operates for 45 minutes each day on average. Find the cost of using the heater for 30 days. (2 marks)

*21 Misc Figure k shows a part of a household electrical wiring plan. The mains cable enters the consumer unit via a kilowatt-hour meter. At the consumer unit, the wires branch out into different circuits.

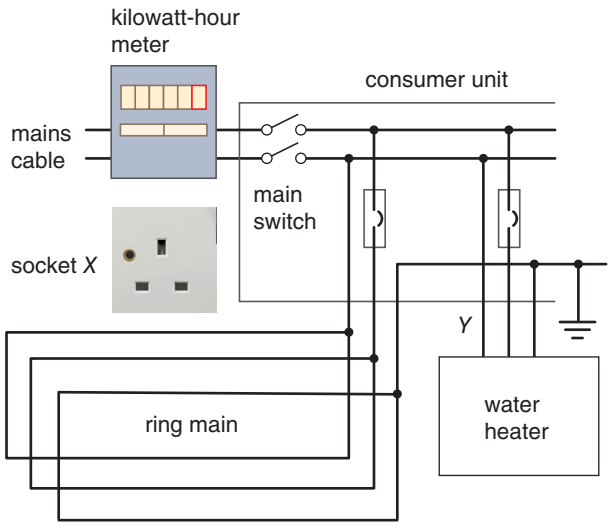


Fig k

(a) Draw wires in Figure k to connect socket *X* to the ring main. (2 marks)

(b) The water heater is rated at '220 V, 2500 W'.

(i) Find the resistance of the water heater. (1 mark)

Ext (ii) What is the peak current at *Y*? (1 mark)

(iii) Which cable, with current rating at 2 A, 6 A, 17 A and 63 A, is the most suitable to be used at *Y*? (2 marks)

(c) The water heater operates 2 hours per day. How much should be paid per day to run the water heater? (1 mark)

**22 Misc Immersion heater *X* is connected to the 220-V mains and is used to boil 2 kg of water at 25 °C. Only 85% of the energy input to *X* becomes the internal energy of the water. The specific heat capacity of water is 4200 J kg^{-1} $°C^{-1}$.

(a) Find the cost of boiling the water. (3 marks)

(b) Immersion heater *Y* is also connected to the mains and is used to boil 2 kg of water at 25 °C. Like *X*, only 85% of the energy input to *Y* becomes the internal energy of the water. The power of *Y* is lower than that of *X*. Compare the cost of boiling the water using *X* and *Y*. Explain. (2 marks)

(c) *X* and *Y* are now connected in series to the mains.

(i) Which heater has a higher power consumption? (3 marks)

(ii) A student suggests that the time needed to boil the water using this set-up will be shorter than using either heater alone. Comment on this idea. (3 marks)

Refer p.95, 146

**23 4.2 An electric travel kettle is designed to operate at 220 V or 110 V with the same power consumption of 2000 W. The circuit of the kettle is shown in Figure l.

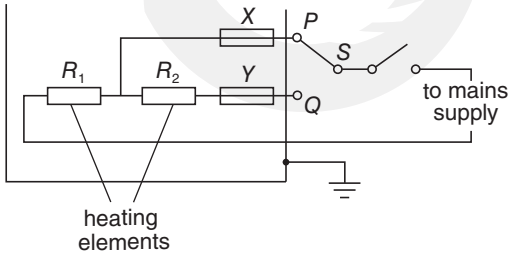


Fig l

The kettle is now connected to the 220-V mains.

(a) To which point, *P* or *Q*, should *S* be connected? (1 mark)

(b) Find the current ratings (10 A, 15 A or 20 A) of fuses *X* and *Y*. (3 marks)

(c) What will happen if *S* is connected incorrectly? (1 mark)

(d) Find the resistances of R_1 and R_2. (3 marks)

Refer p.148

See p.183 for the Exam Report.

24 HKDSE 2018 Paper 1B Q8 4.2

(a) Figure m shows the schematic diagram of an electric heater which can operate in two modes, namely, 'heating' and 'keeping warm'. The heating elements of resistances 4*R* and *R* are connected to the mains supply via a 3-way switch with its two poles tied together. That is, both poles can be connected to one of the three pairs of terminals *X*, *Y* or *Z*.

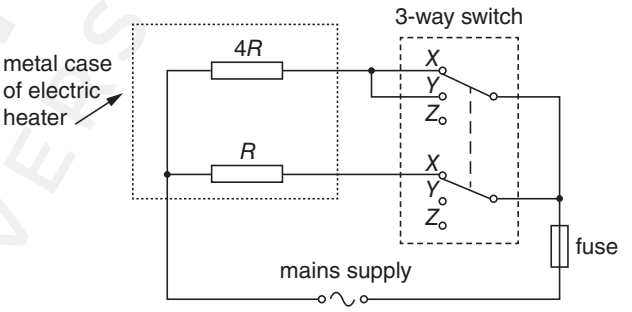


Fig m

(i) To which pairs of terminals, *X*, *Y* or *Z*, should the switch connect to when the heater is in 'heating' mode? (1 mark)

The power consumed by the heater in 'heating' mode is 800 W.

(ii) Calculate the current drawn from the 220 V mains supply when the heater is in 'heating' mode. (2 marks)

(iii) Find the power consumed by the heater in the mode of 'keeping warm'. (3 marks)

(b) Figure n shows a simplified domestic circuit connected to an electrical appliance via a fuse, a meter *M*, a residual current circuit breaker (RCCB) and a switch.

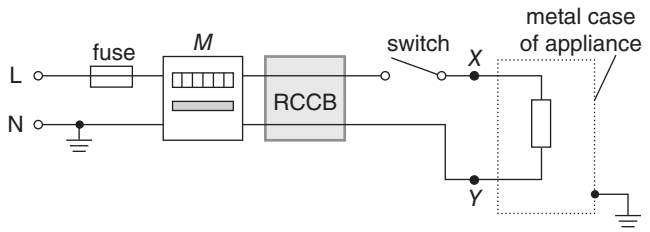


Fig n

(i) What physical quantity does the meter M record? Electrical energy (consumed) (1 mark)

(ii) An RCCB is a kind of safety device that cuts off the supply automatically whenever there is a small difference between the currents in the live (L) and neutral (N) wires. State, in each of the following situations, which device(s) will respond (i.e. the fuse blows and/or the RCCB cuts off the supply).

(1) A short circuit occurs between points X and Y. (1 mark)

(2) A short circuit occurs between point Y and the metal case of the appliance. (1 mark)

25 HKDSE 2019 Paper 1B Q8

4.2

Figure o shows a household electrical wiring circuit. The mains cable (containing live wire L and neutral wire N) is connected to a consumer unit via a kilowatt-hour meter M. At the consumer unit, the wires branch out into a number of parallel circuits.

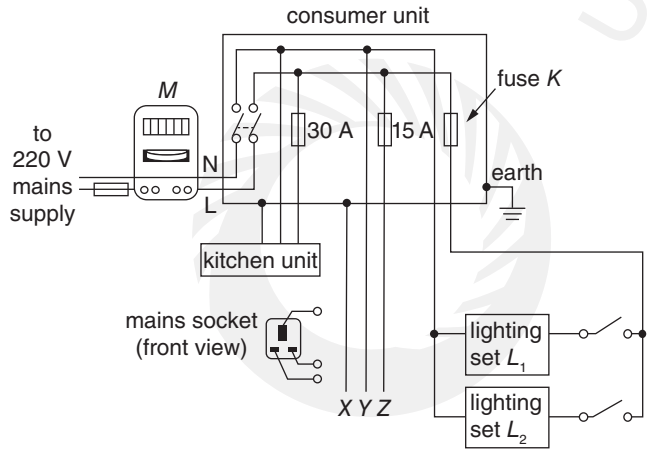


Fig o

(a) Indicate on Figure o how the mains socket should be connected to wires X, Y and Z. (1 mark)

(b) Lighting sets L_1 and L_2 of power ratings 300 W and 450 W respectively are connected in parallel to the branch with fuse K.

(i) State one advantage of connecting L_1 and L_2 in parallel instead of in series to the branch. (1 mark)

(ii) If fuses marked 3 A, 5 A, 10 A and 13 A are available, which one is the most suitable to be fuse K? Explain your choice. (3 marks)

(c) The kitchen unit includes the following electrical appliances:

	Rating	Effective time of operation at rated value per day
A refrigerator	220 V, 500 W	8 hours
An electric kettle	220 V, 2000 W	0.5 hour
An induction cooker	220 V, 3000 W	2 hours

Table a

How much should be paid per day to run these appliances if 1 kW h of electrical energy costs \$0.9? (3 marks)

Q24 Exam Report: A considerable number of candidates wrongly chose terminals Y in (a)(i) for the 'heating' mode. Even for those who chose terminals X and correctly found the current drawn from the supply in (a)(ii), some wrongly employed this total current to find the power consumed in the mode of 'keeping warm' in (a)(iii). In (b)(i), many thought that the meter M was to record voltage, current or power. (b)(ii) was in general well answered although some candidates did not attempt this part possibly as it was too challenging.

Q25 Exam Report: The question tested candidates' knowledge and understanding on domestic circuits. The overall performance was good. Most answered part (a) correctly though some drawings of the connecting wires were quite messy. In (b)(i), many candidates knew the advantage of connecting L_1 and L_2 in parallel, however, some gave incomplete or wrong answers like 'both can work under the same voltage' or 'when one lighting set failed to work due to a short circuit, the other can still work'. Although most managed to find the working currents of the two lighting sets in (b)(ii), a few wrongly thought that the fuse value should be greater than either currents but not the total. Part (c) was well answered.

Know more

Sinusoidal a.c. (p.171)

For a sinusoidal a.c. (Fig a), the value of I can be expressed as:

$$I = I_p \sin \frac{2\pi t}{T}$$

where I_p is the peak value of current and the angle of the sine function is measured in radians.

The power dissipated by a pure resistor R opeated by this a.c. at time t is given by:

$$P = I^2R = \left(I_p \sin \frac{2\pi t}{T}\right)^2 R = I_p^2 R \left(\sin \frac{2\pi t}{T}\right)^2$$

As $P_p = I_p^2 R$,

$$\therefore \quad P = P_p \left(\sin \frac{2\pi t}{T}\right)^2$$

Figure b shows how the power P vary with time t. Although the value of $\sin \frac{2\pi t}{T}$ can be negative, the values of $\left(\sin \frac{2\pi t}{T}\right)^2$ is always non-negative.

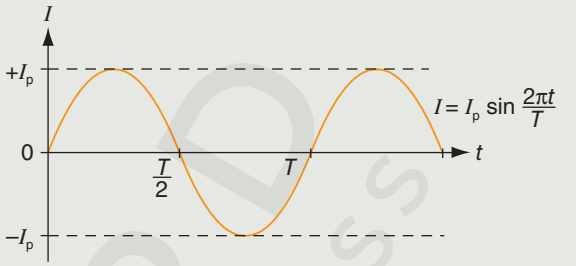


Fig a *Graph of I against t, fluctuating between $-I_p$ and $+I_p$.*

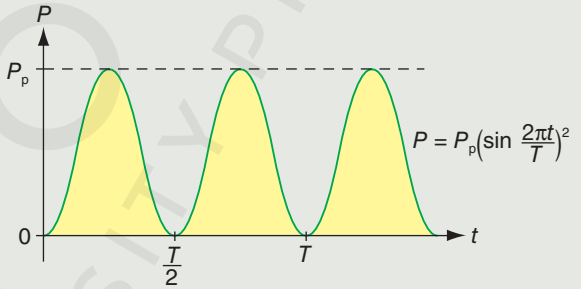


Fig b *Graph of P against t, fluctuating between 0 and P_p.*

Consider the area of the shaded regions bounded by the power curve and the t-axis in Figure c. Indeed, the regions above the horizontal line $P = \frac{P_p}{2}$ (areas in pink in Figure c) can just fit into the empty regions below $P = \frac{P_p}{2}$. This means that the area under the power curve is equal to the area under the horizontal line $P = \frac{P_p}{2}$.

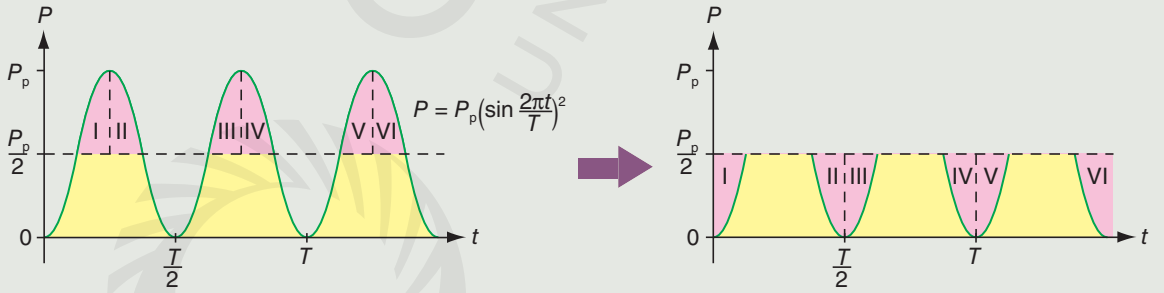


Fig c *Obtain the mean power $\overline{P}$ from the P–t graph.*

Therefore, the average power $\overline{P}$ provided by the sinusoidal a.c. can be expressed as:

$$\overline{P} = \frac{1}{2}P_p \Rightarrow I_{rms}^2 R = \frac{1}{2}I_p^2 R \Rightarrow I_{rms} = \frac{I_p}{\sqrt{2}}$$

By expressing $P = \frac{V^2}{R}$, the r.m.s. value and the peak value of voltage can be related similarly.

$$V_{rms} = \frac{V_p}{\sqrt{2}}$$

Note that these relations are **valid for sinusoidal a.c. only**.